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(54) **PACKING MACHINE AND METHOD FOR OPERATION**

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(58) **Field of Classification Search**
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See application file for complete search history.

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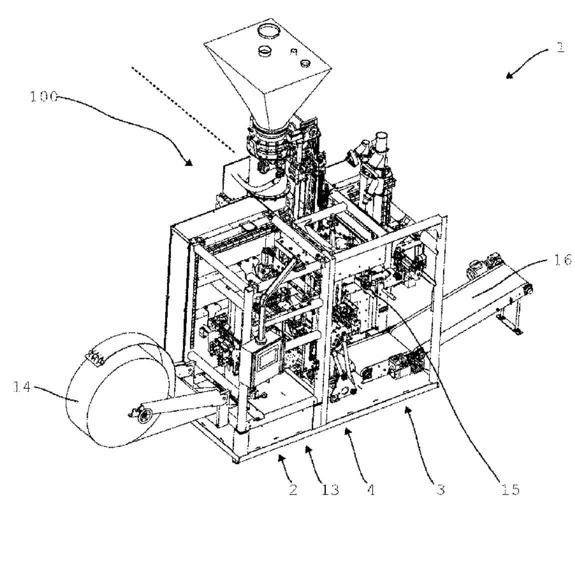
(57) **ABSTRACT**

A packing machine for filling product into bags including a bag supply, a filling module with a filling nozzle and a bag removal. In this case, the bag supply and the bag removal are arranged in a line and the filling module is arranged pulled out of the line. In a method for operating such a packing machine, a bag from the bag supply is fed out from the line to the filling module and after filling is guided back into the line and fed to the bag removal.

16 Claims, 16 Drawing Sheets



(51) **Int. Cl.**

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B65B 31/02	(2006.01)
B65B 37/02	(2006.01)
B65B 39/06	(2006.01)
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B65B 51/10	(2006.01)
B65B 55/24	(2006.01)
B65B 59/04	(2006.01)
B65B 65/00	(2006.01)
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(52) **U.S. Cl.**

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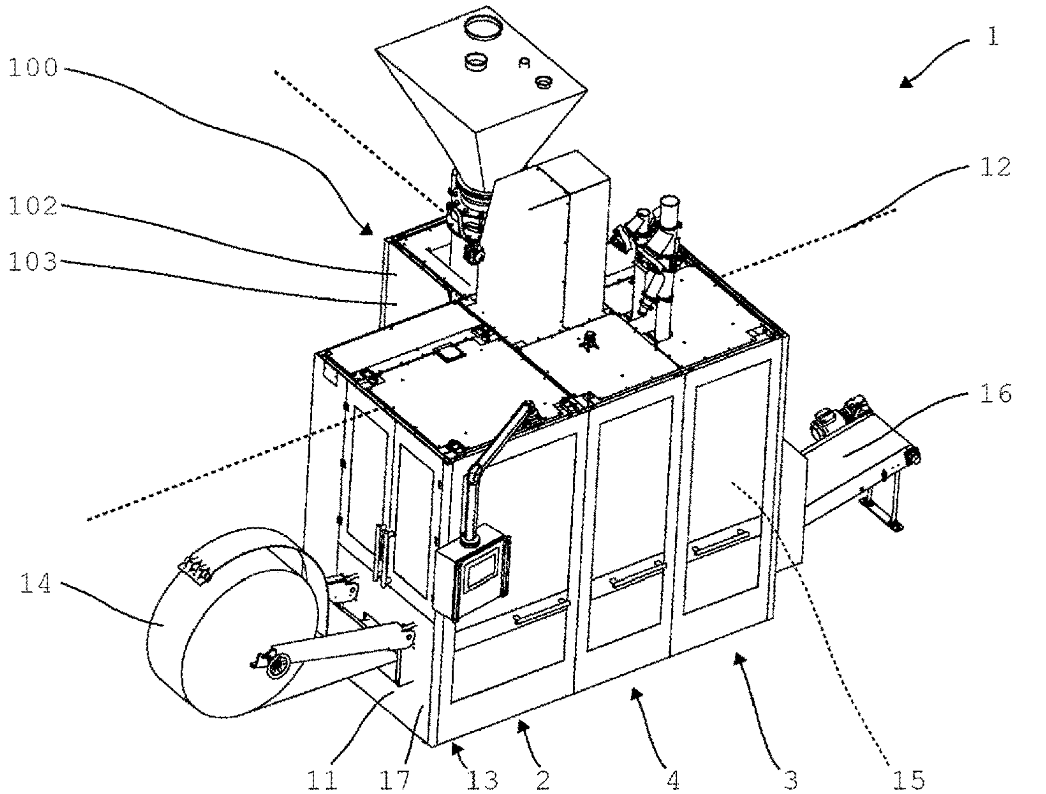


Fig. 1

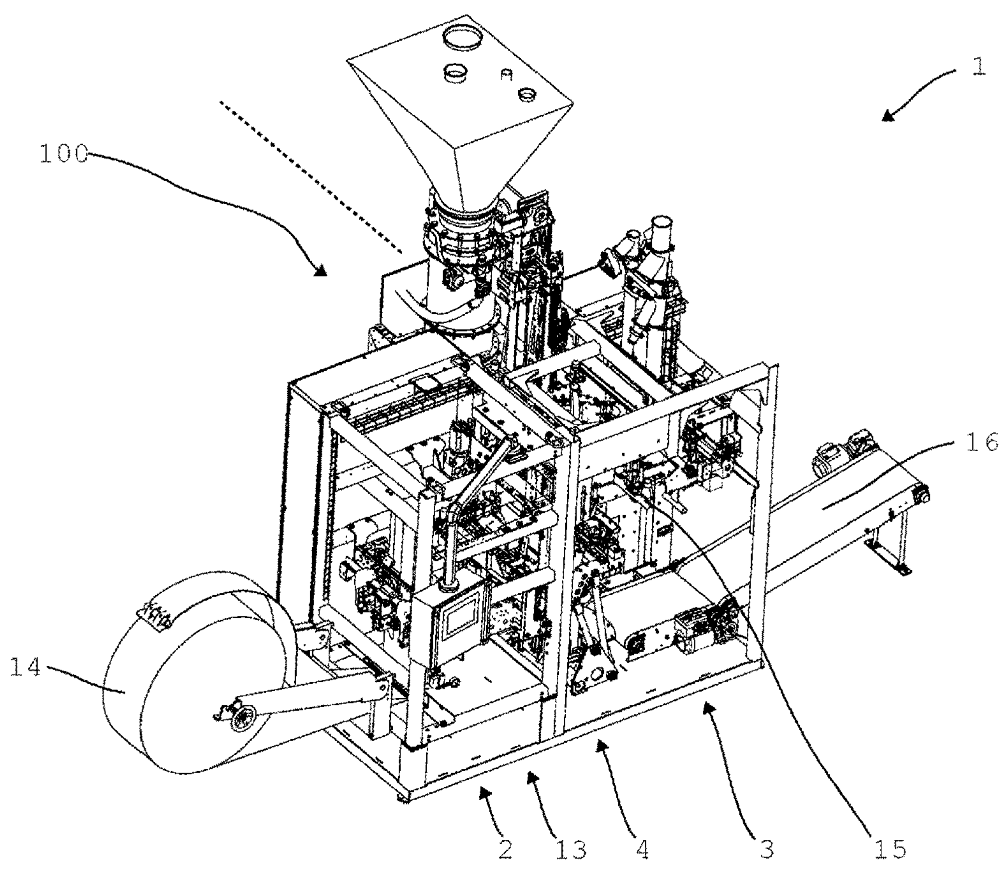
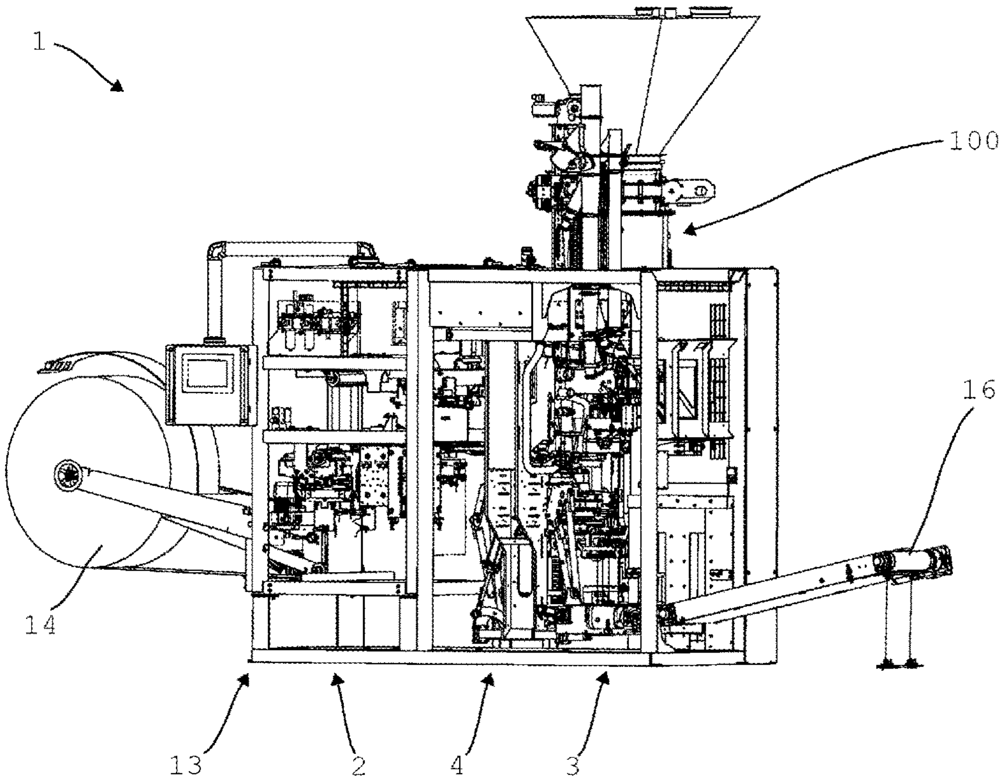


Fig. 2



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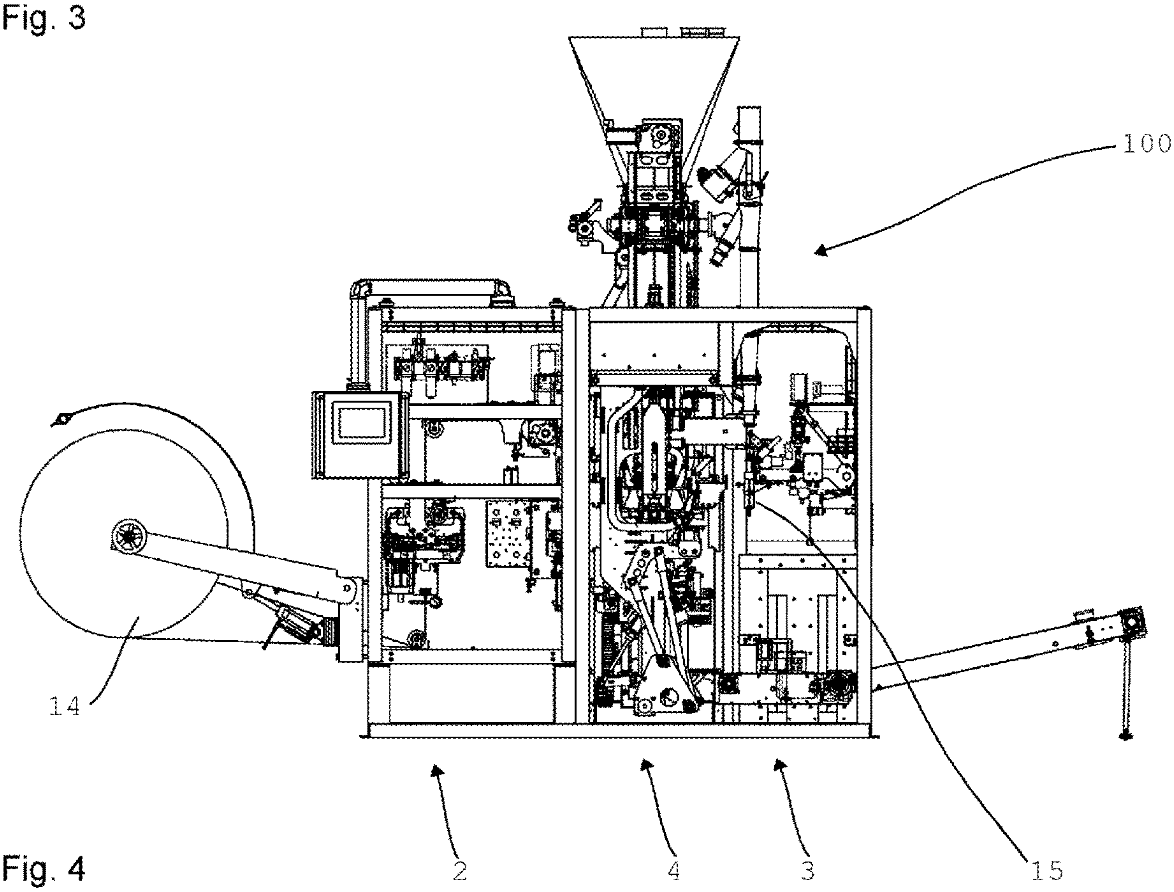
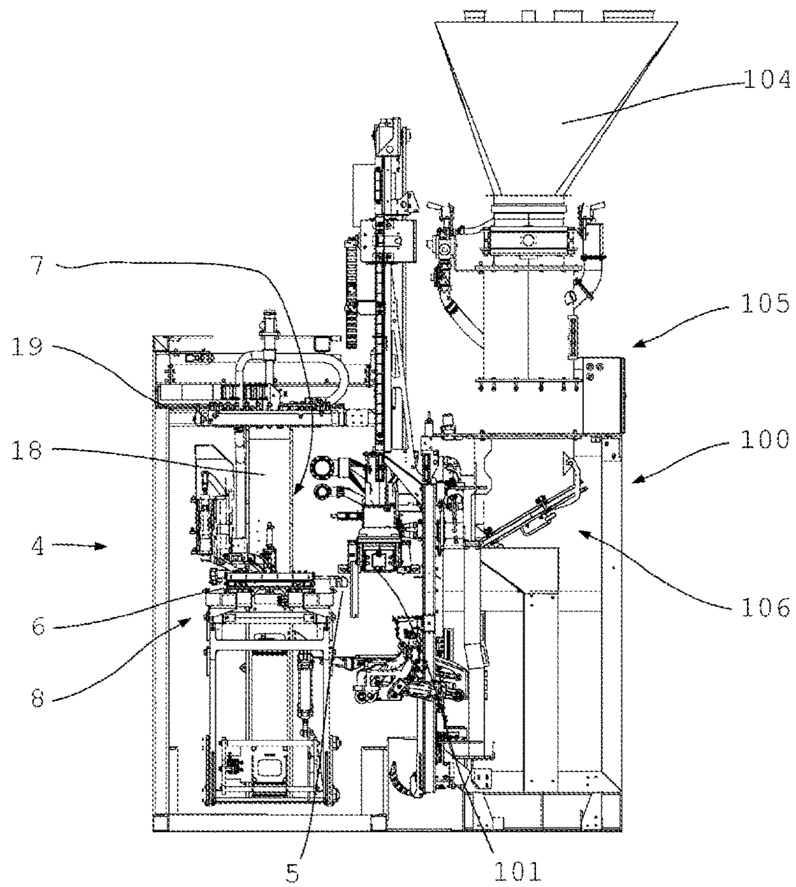


Fig. 4



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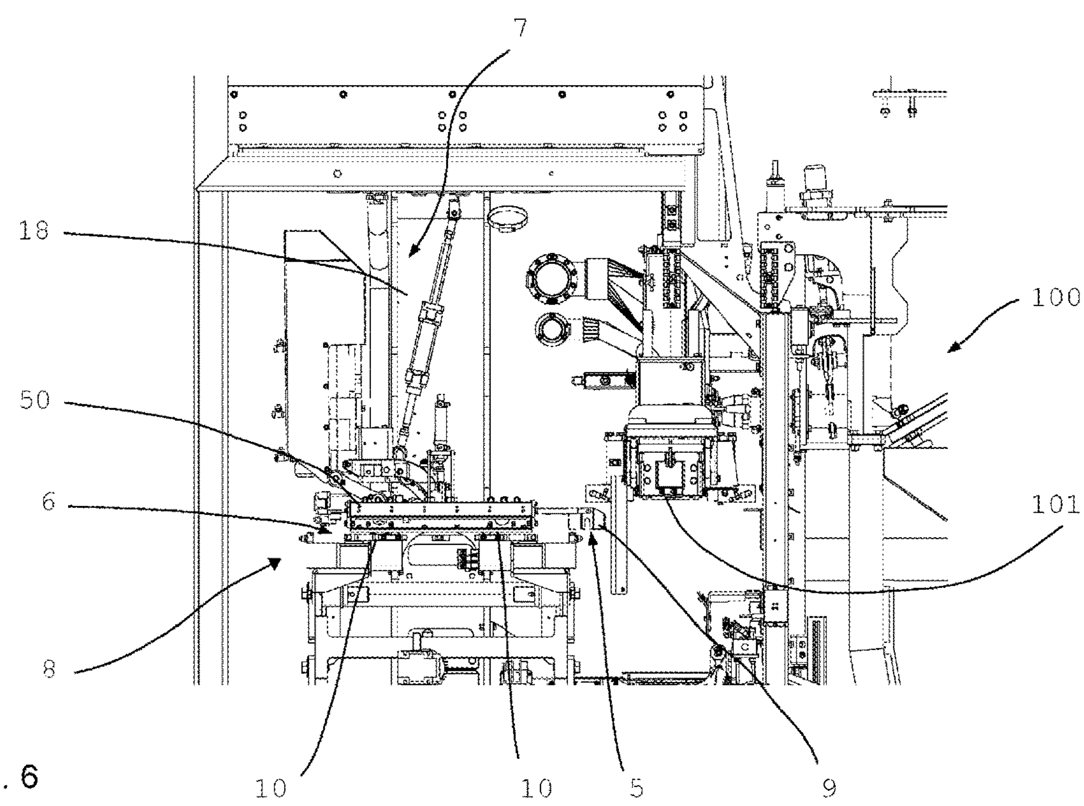

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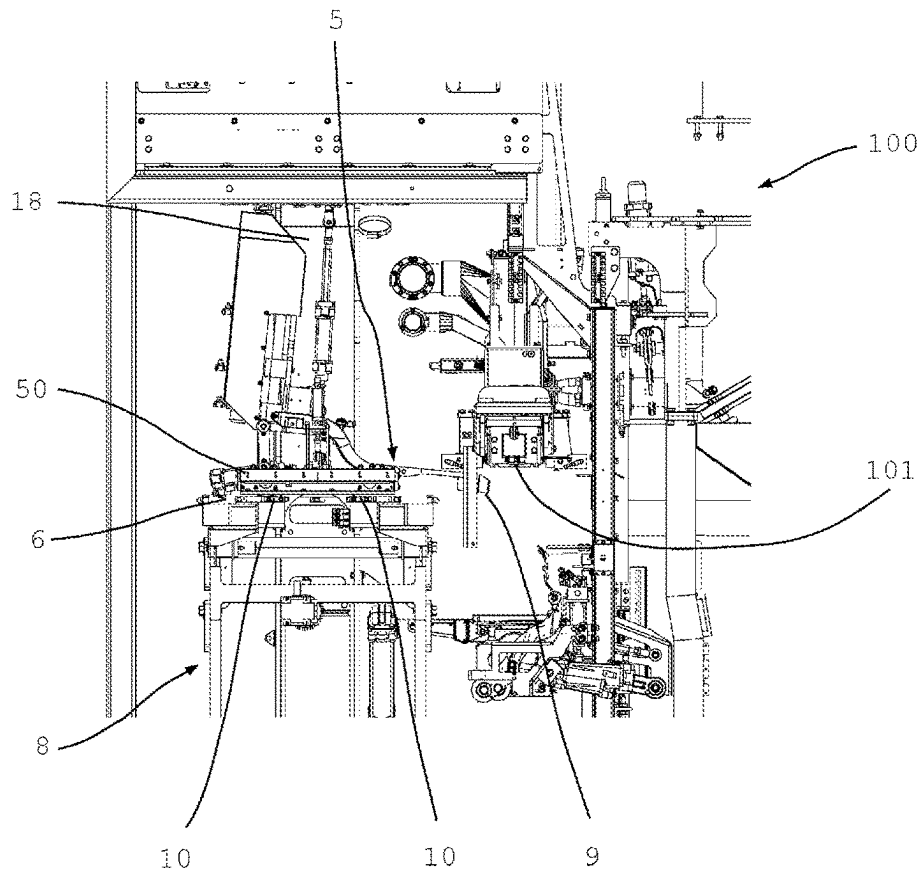

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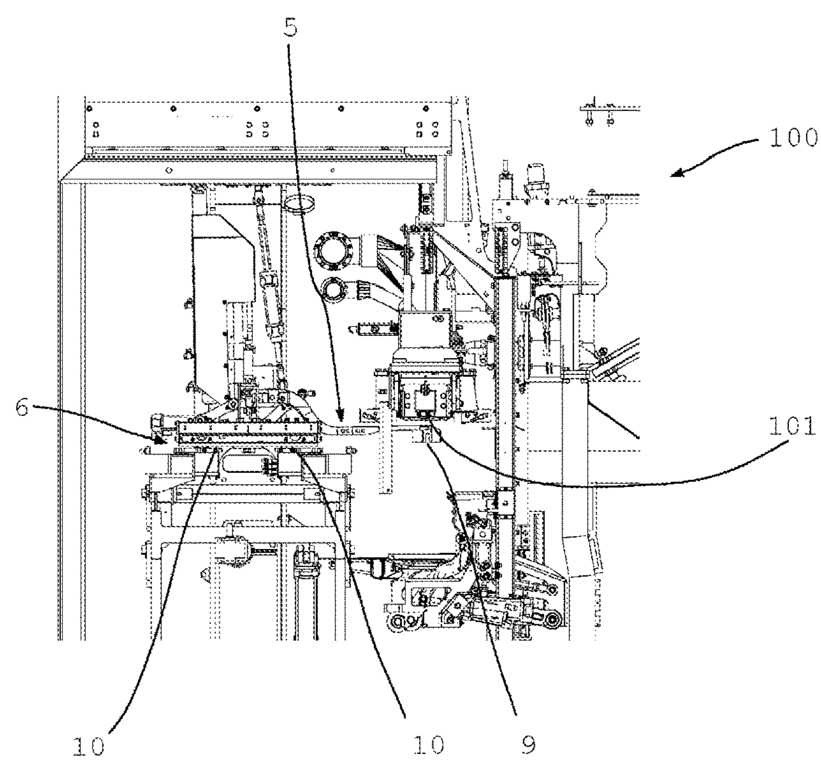

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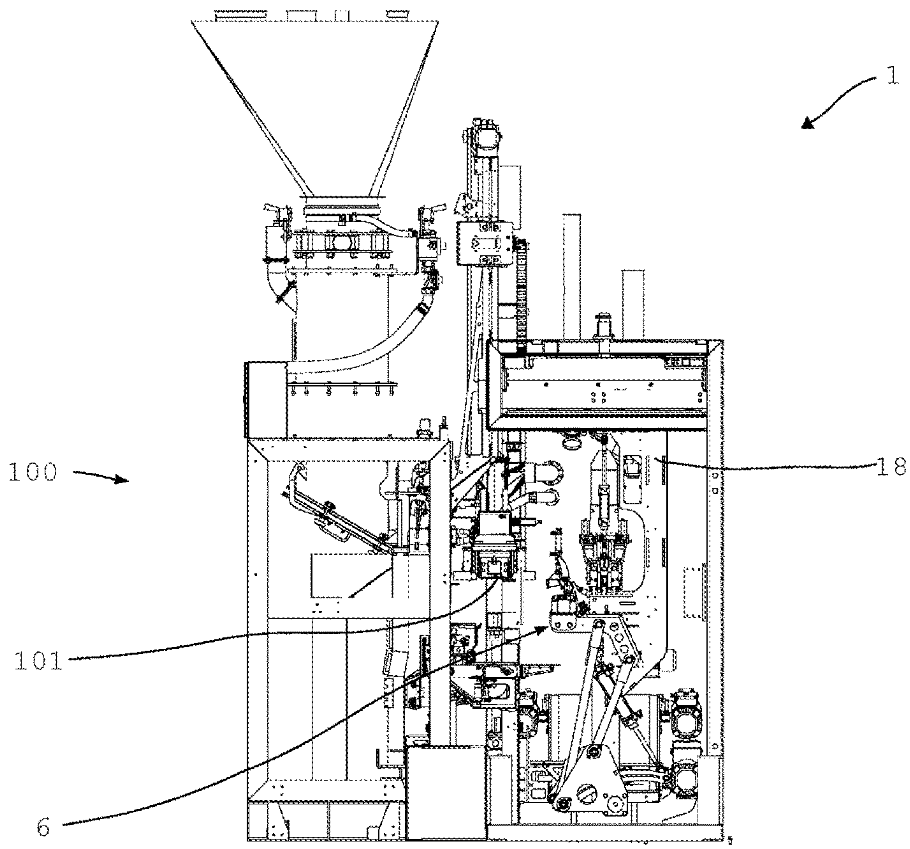

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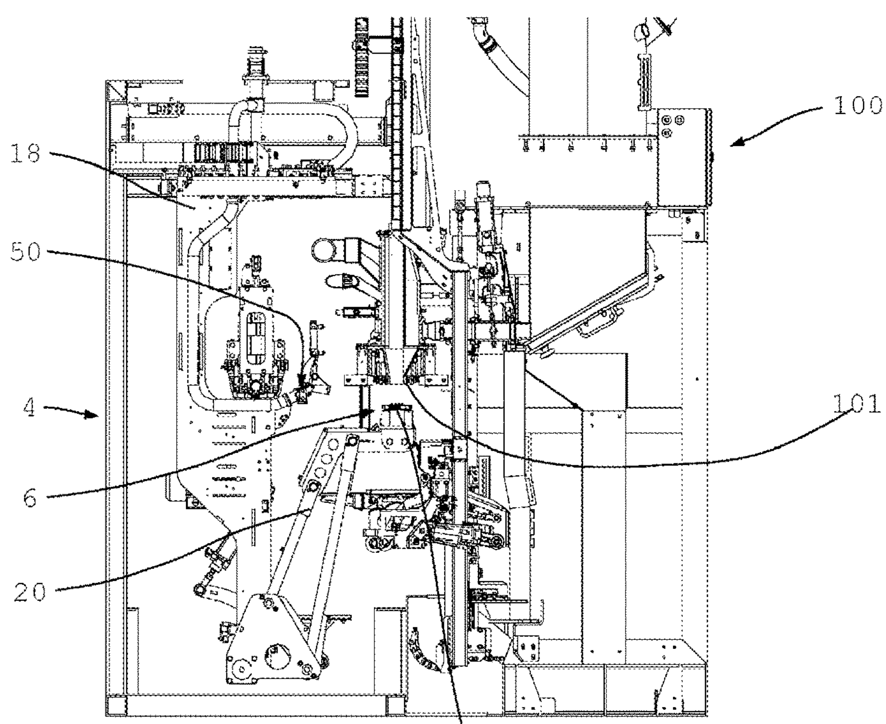

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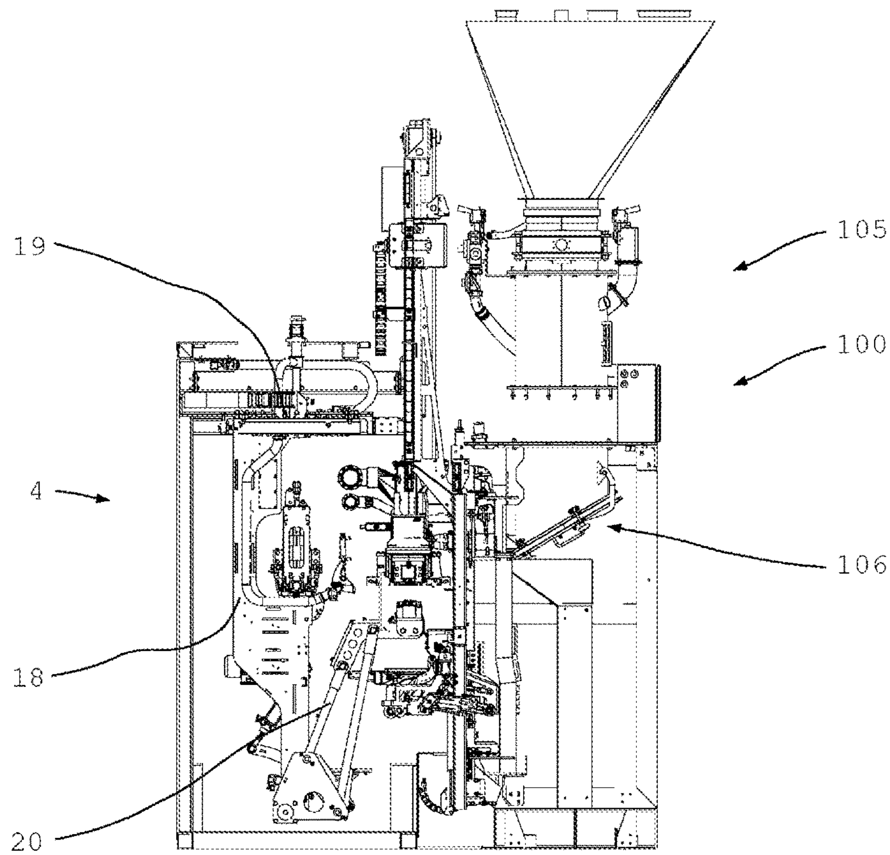

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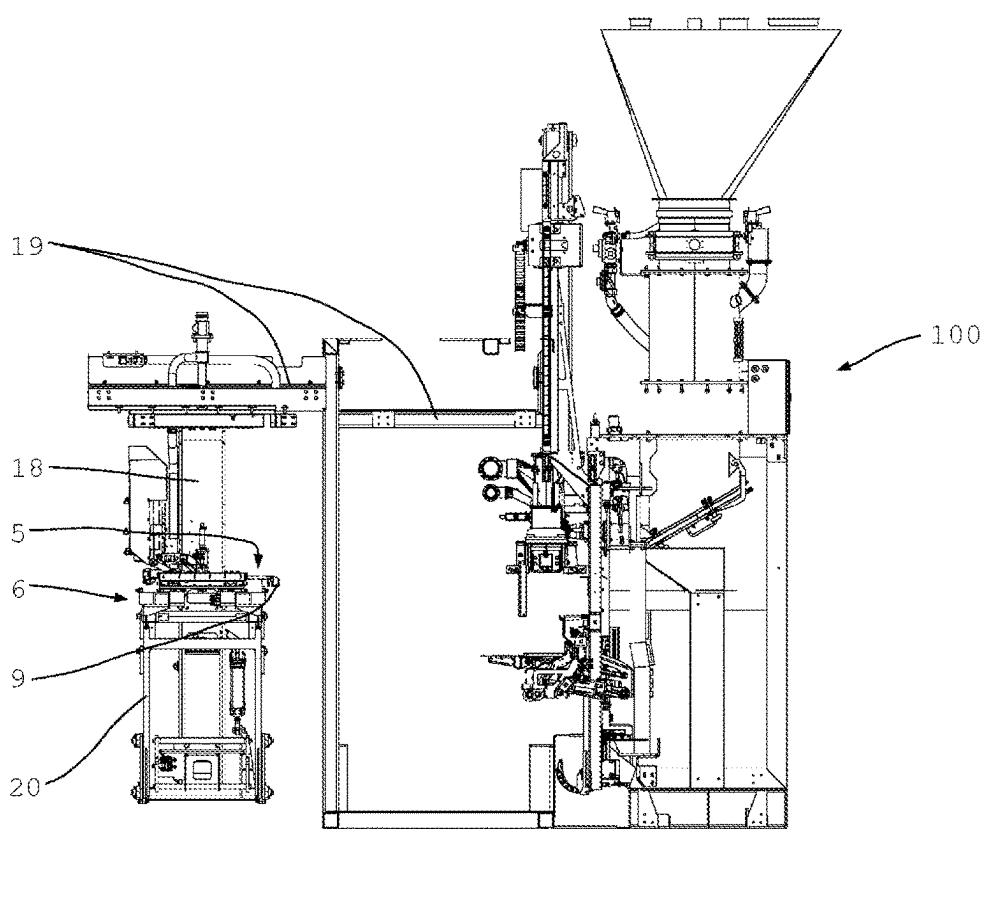

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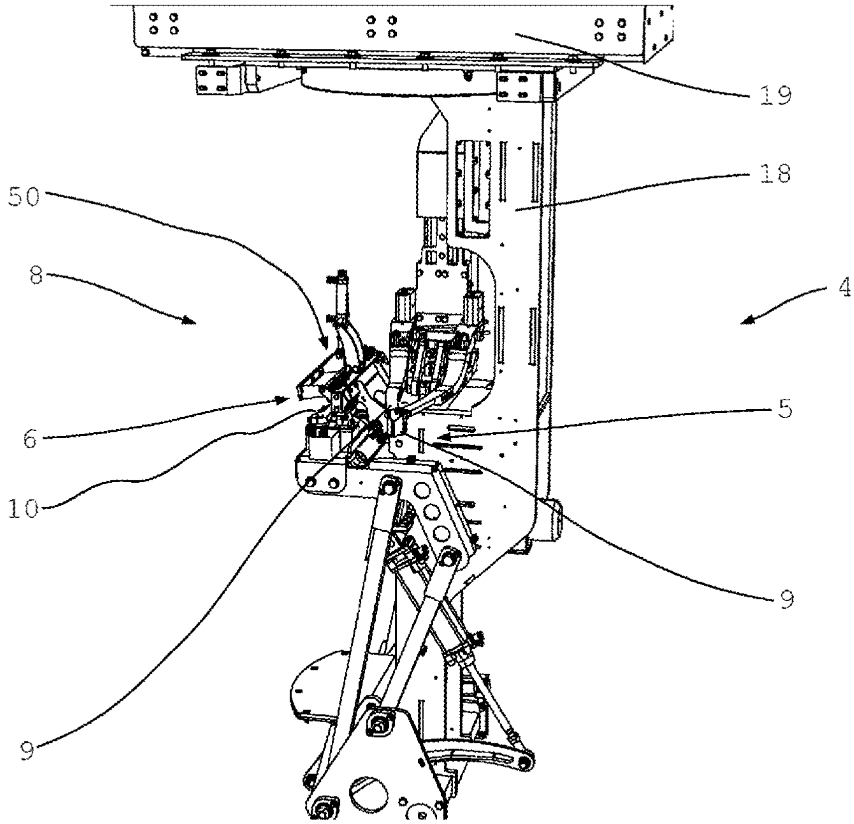

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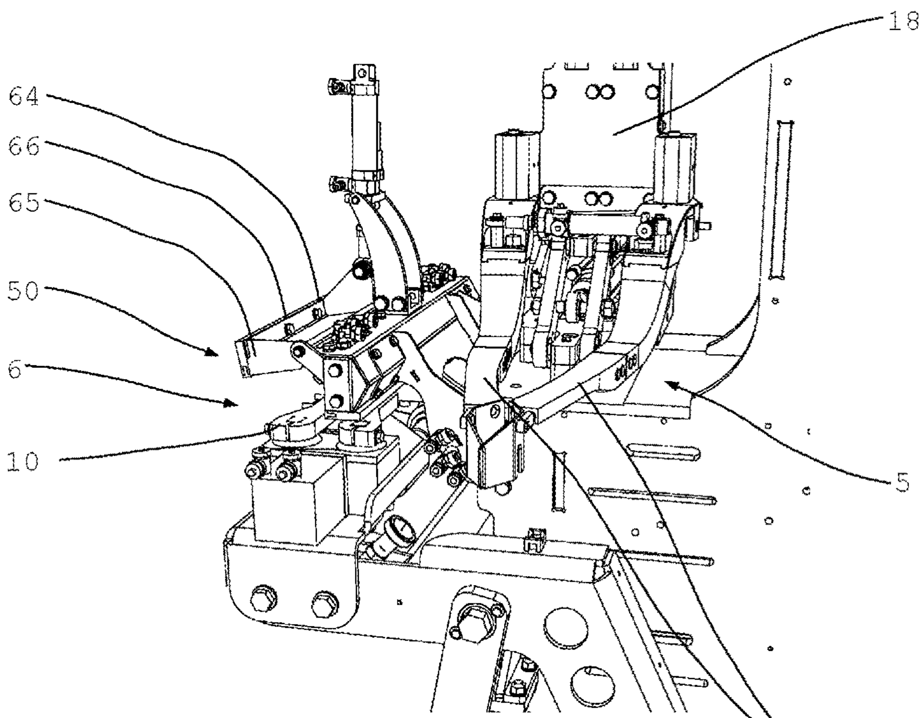

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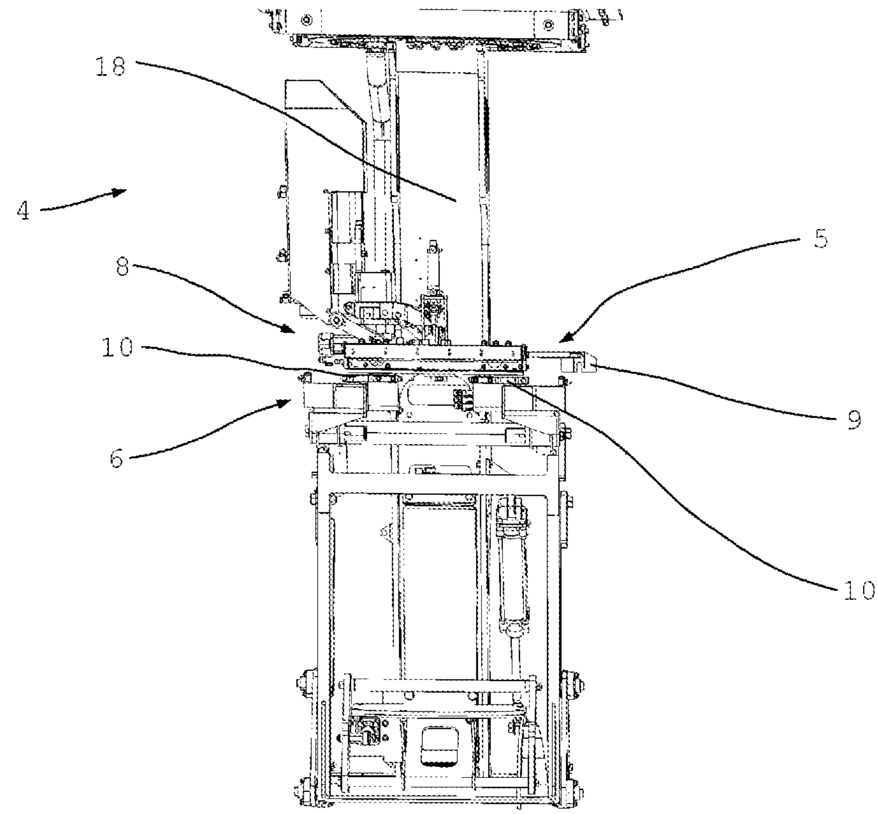

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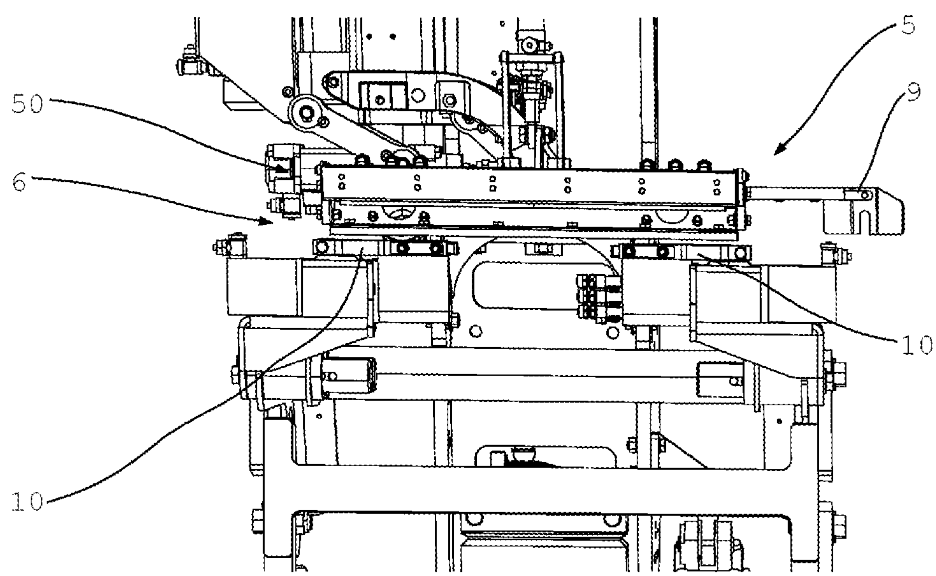

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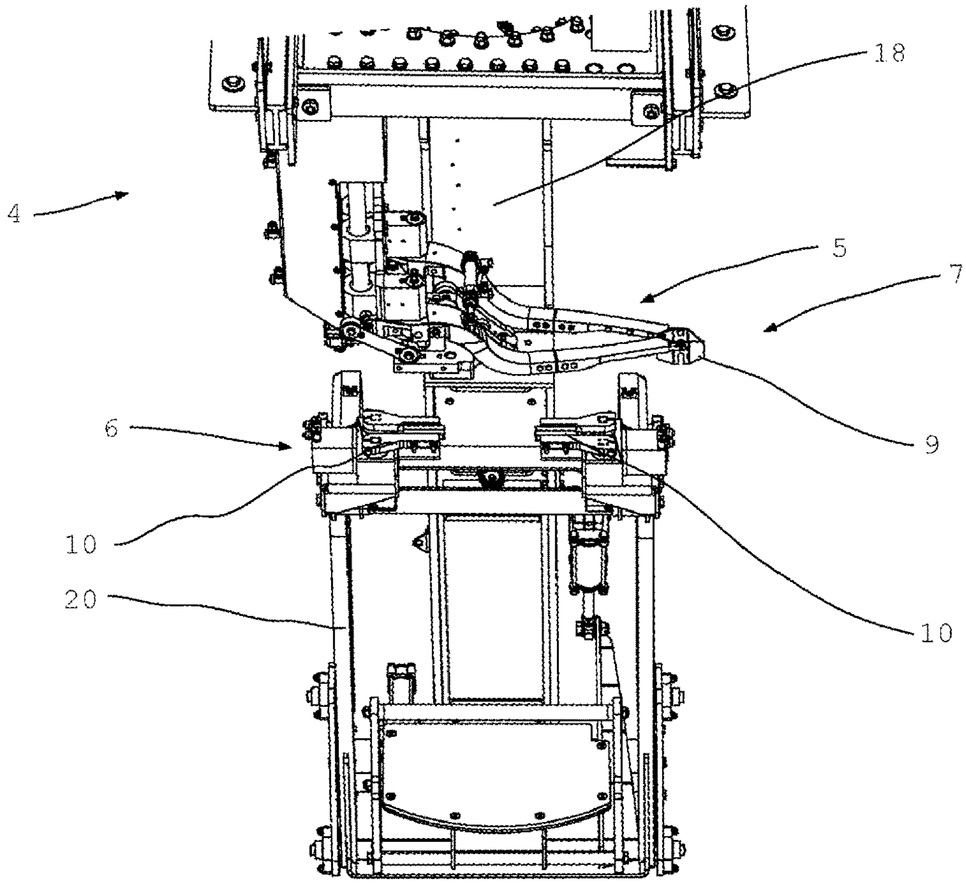

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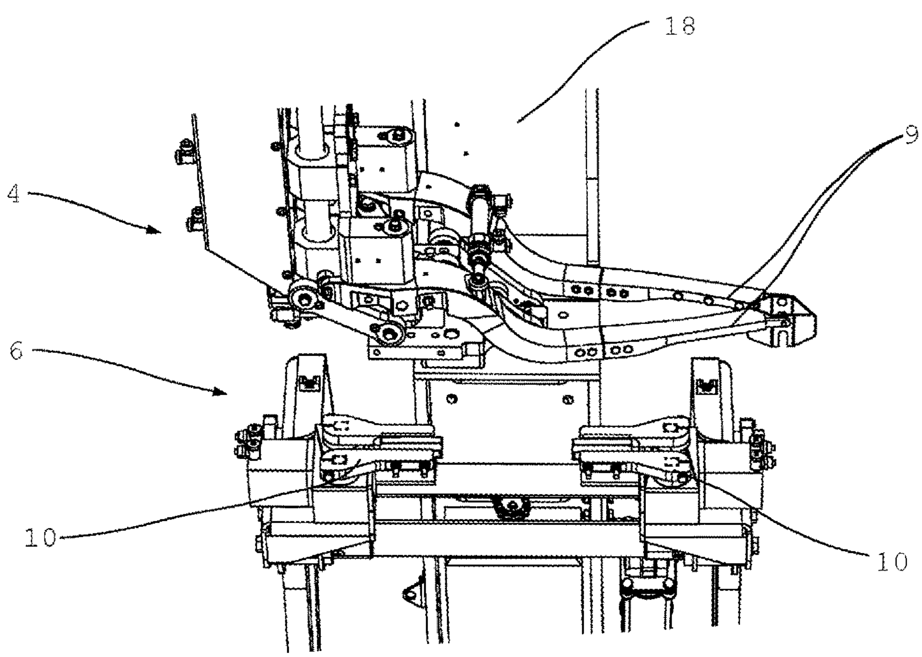

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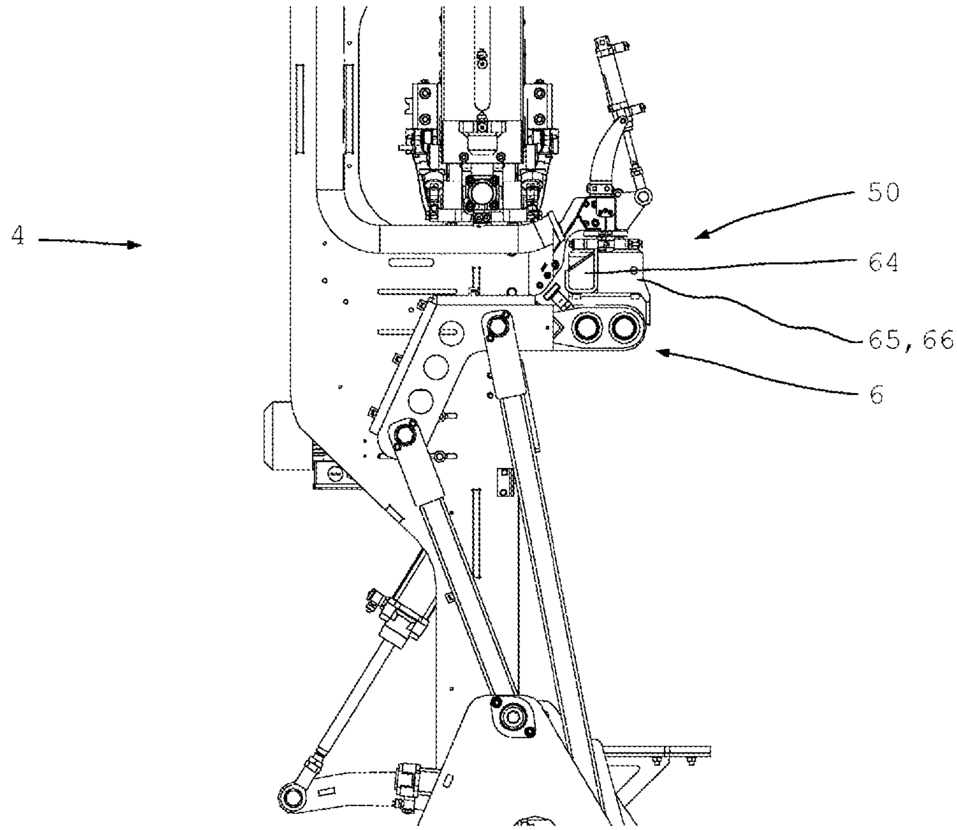

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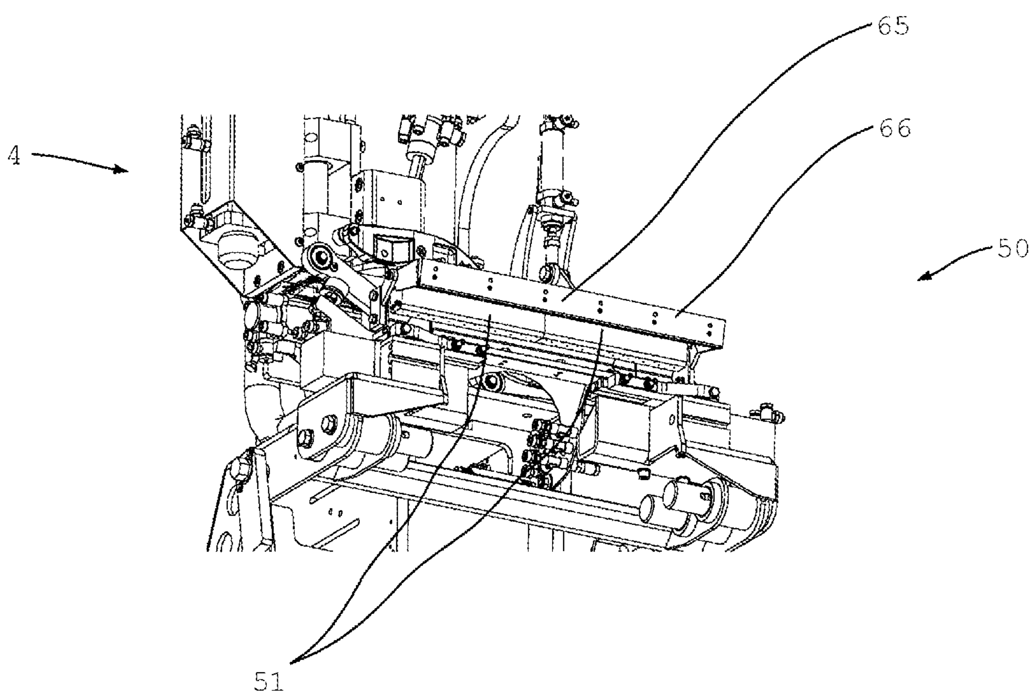

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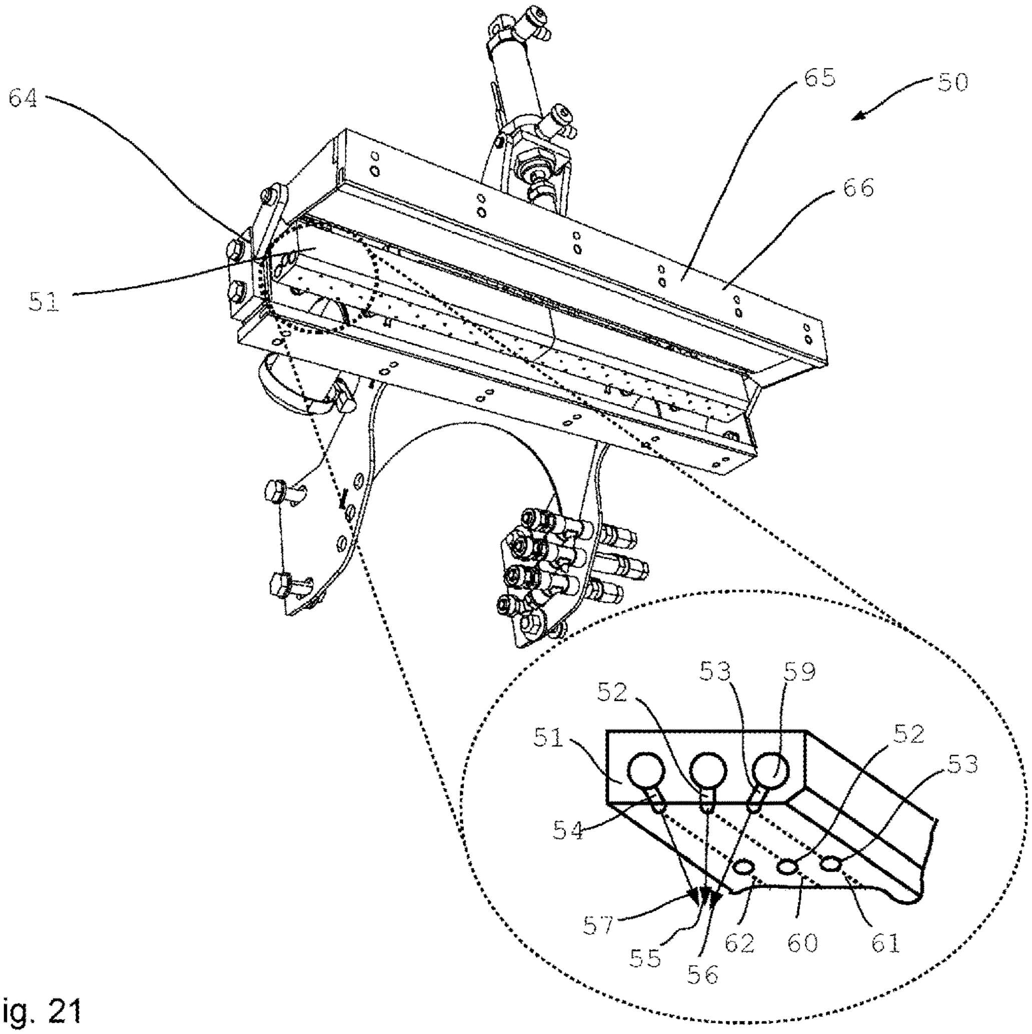

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Fig. 21

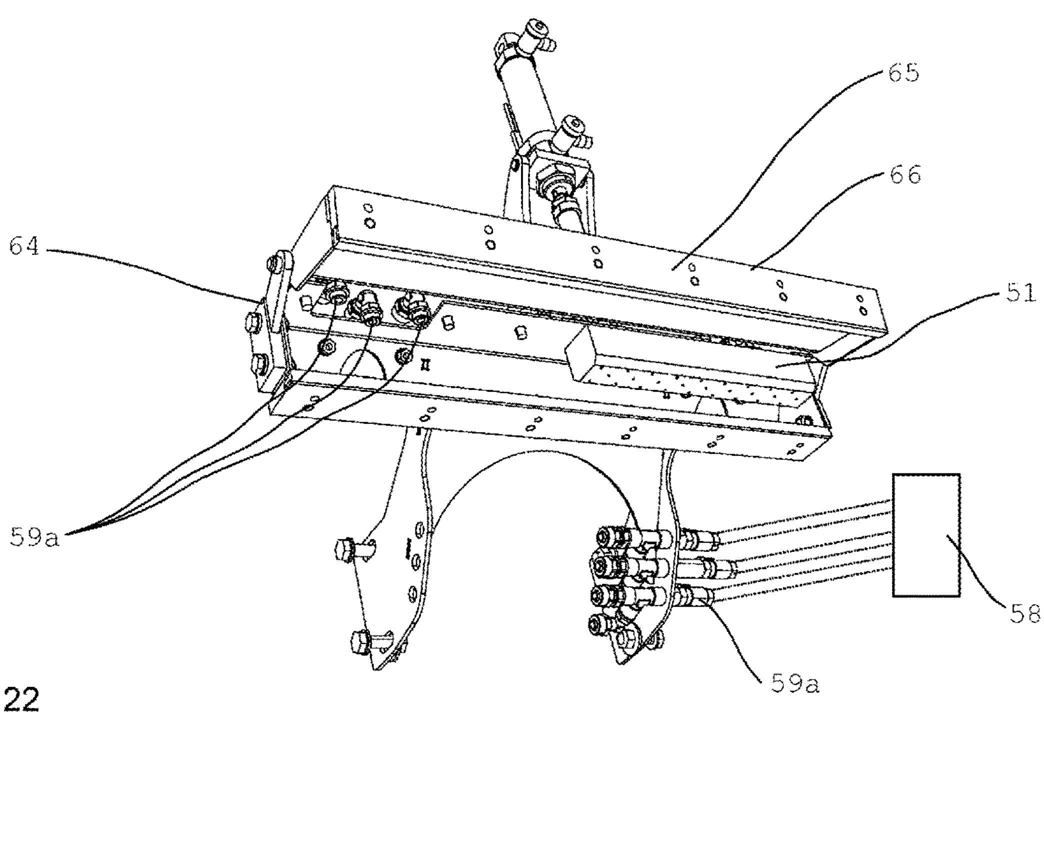

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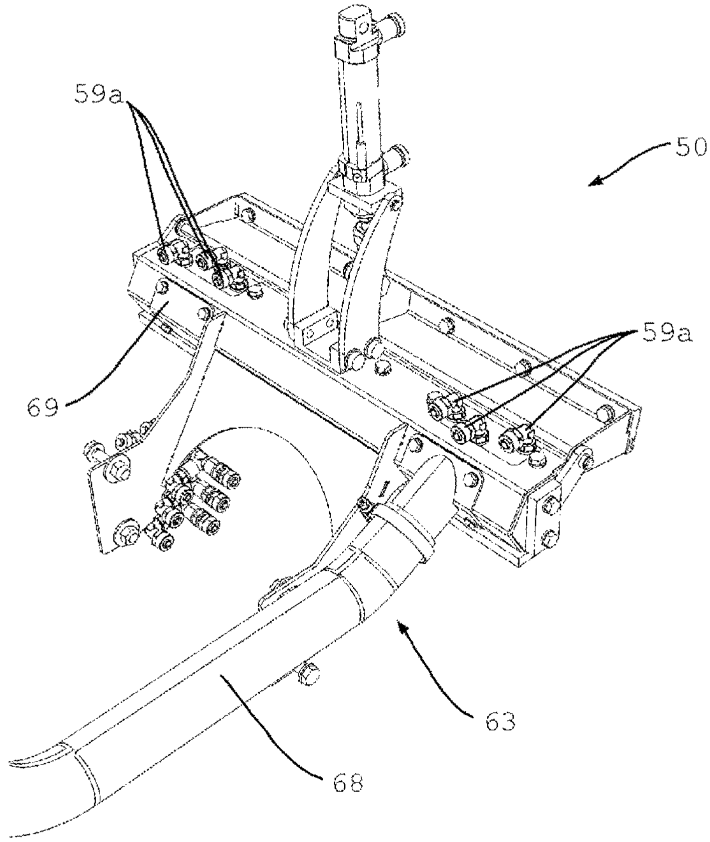
Fig. 22



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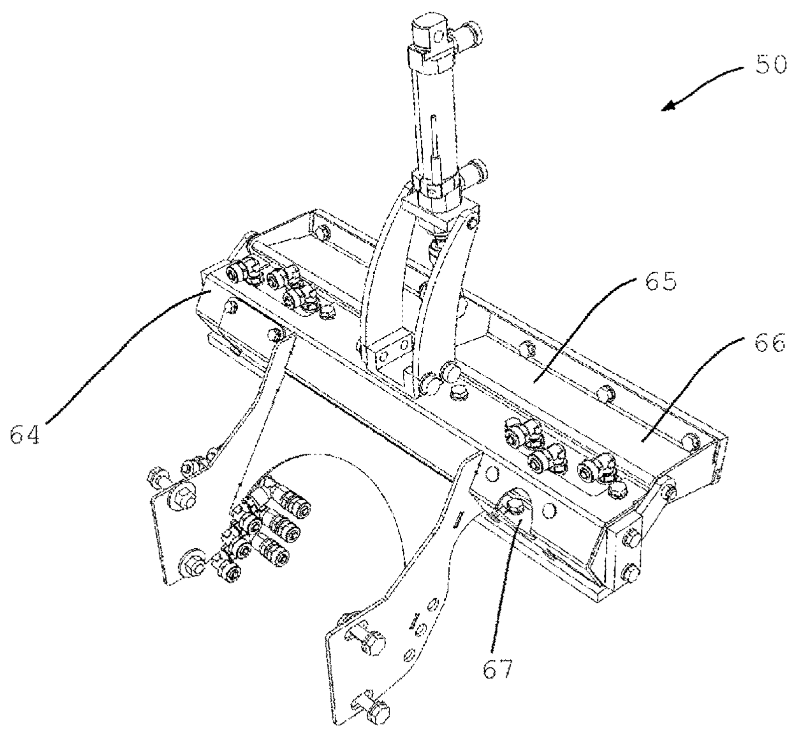

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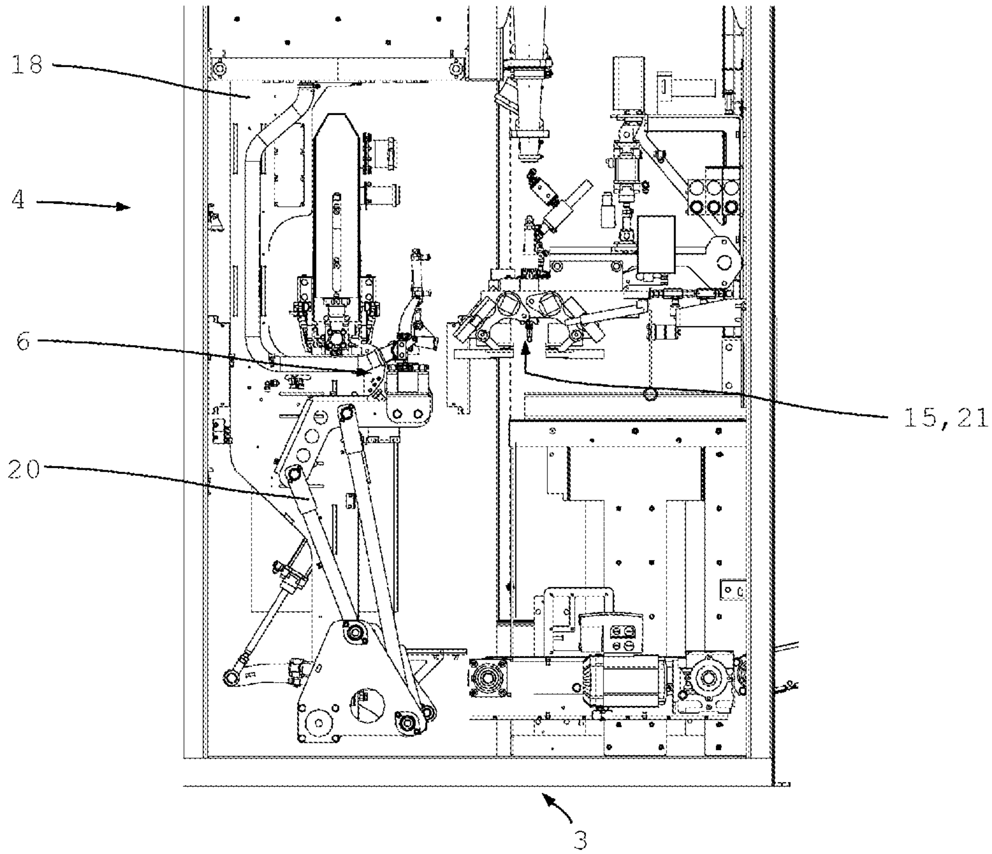

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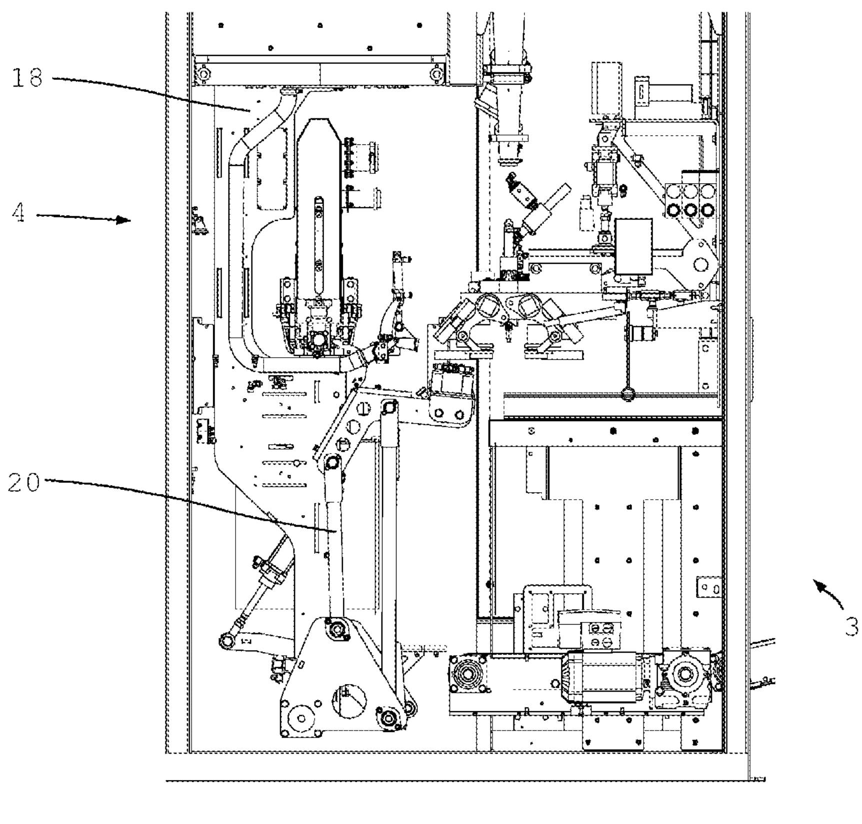
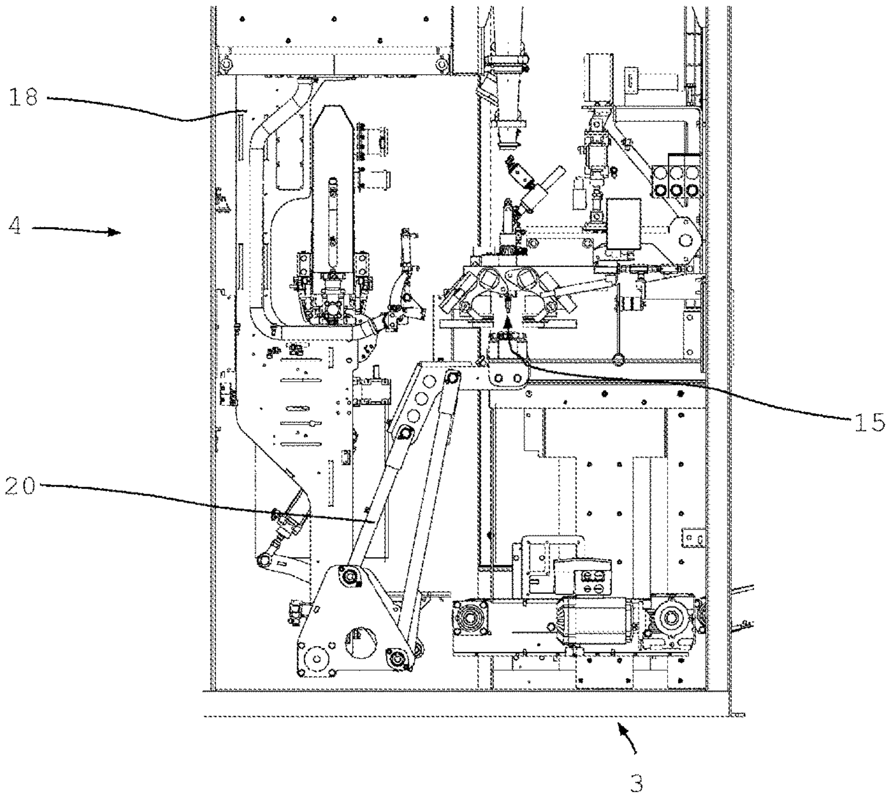


Fig. 26



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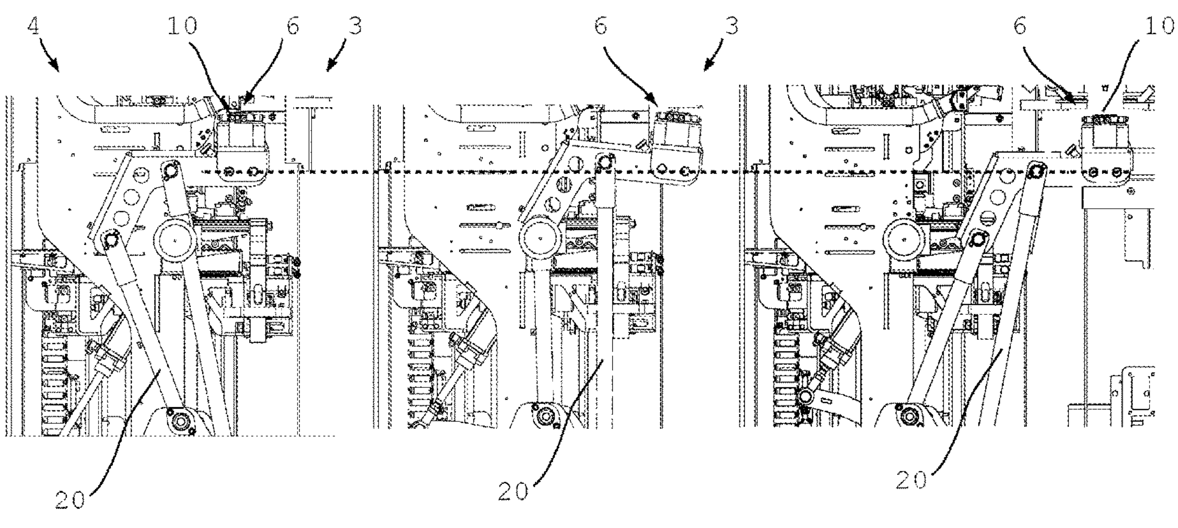
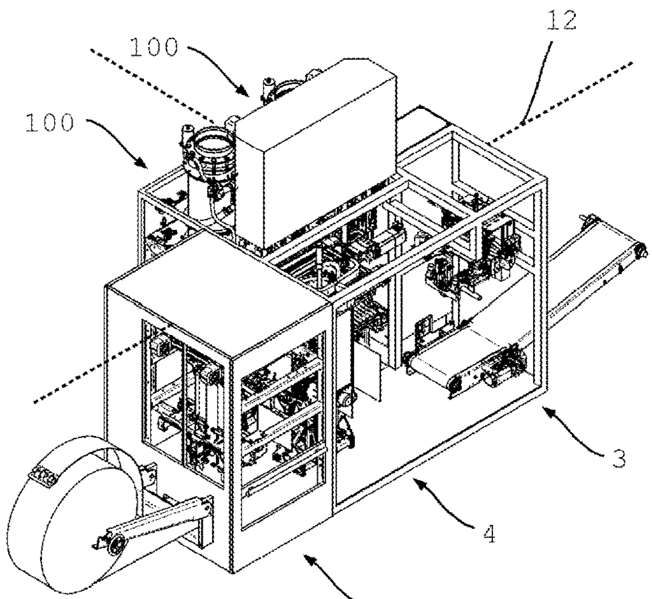


Fig. 28



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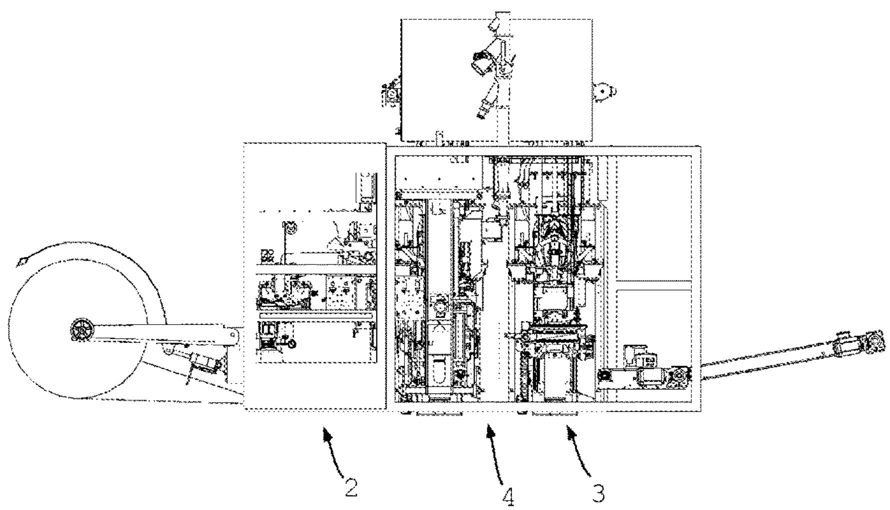
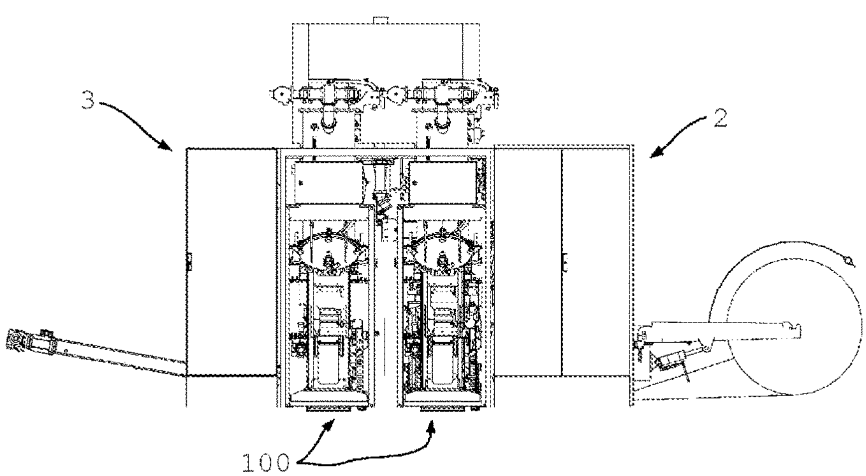


Fig. 30



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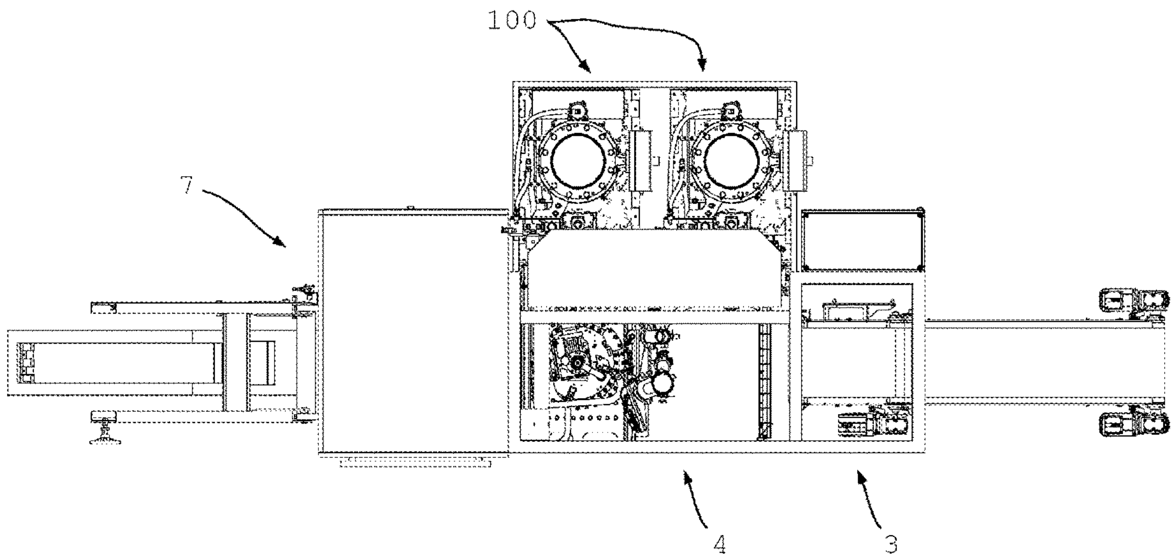

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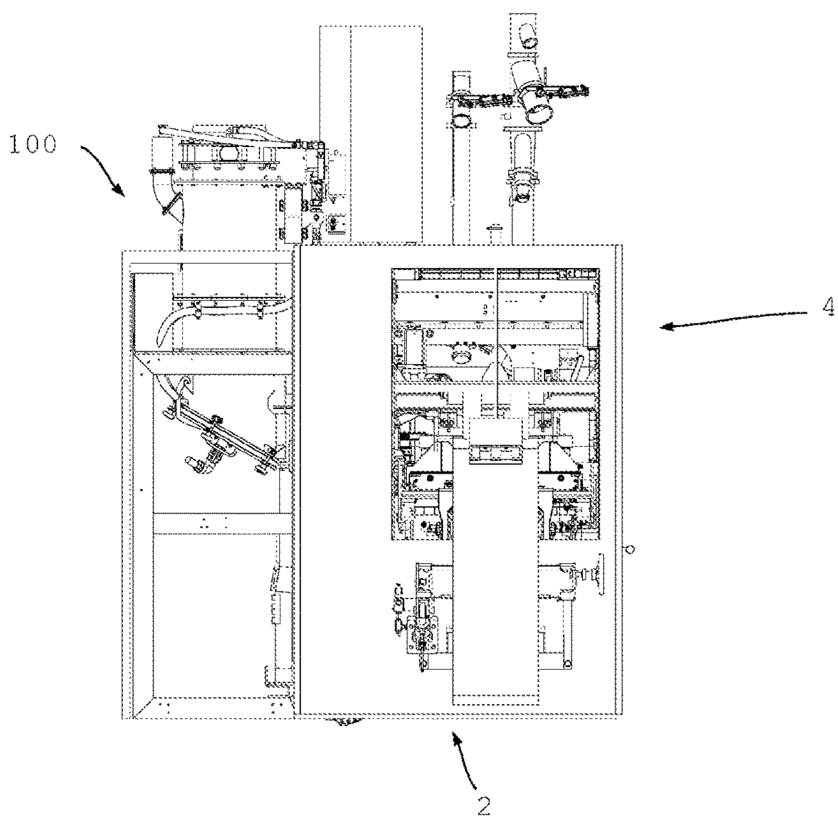

Fig. 33

PACKING MACHINE AND METHOD FOR OPERATION

BACKGROUND

The present invention relates to a packing machine for filling bulk materials into bags comprising at least one bag supply, at least one filling module with at least one filling nozzle and at least one bag removal. The present invention additionally relates to a method for operating such a packing machine.

Various installations and methods are known in the prior art for filling bulk materials into containers, in particular for filling pourable and/or free-flowing products into bags.

In this case, there are, inter alia, various and in some cases, special solutions for filling open bags and valve bags. The performance also plays a major role in the type of installation and in the filling method.

In this case, for example, rotating packing installations or packing machines have become known which comprise a plurality of filling modules arranged distributed over the circumference on a type of carousel each having a filling nozzle. For filling the bags a bag is placed or shot onto a filling nozzle at a specific location. During rotation of the packing machine the bag is filled, compacted depending on the design, closed and transferred to another position.

Also substantially linearly arranged packing installations or packing machines have become known in which a bag to be filled is taken from a stack of bags, for example, transferred to a filling module and filled and then removed.

Such installations usually function reliably and can be selected from a plurality of possible configurations as required and demanded.

However, there is still the need for alternative solutions since the known packing machines do not always meet the existing available space and frequently cannot be flexibly adapted to the product to be filled. Also maintenance is expensive in some cases since the known packing installations usually have a very compact and partially installed structure in order to keep the space requirement as low as possible.

It is therefore the object of the present invention to provide a packing machine which can be used flexibly or adapted in a modular manner for example.

SUMMARY

The packing machine according to the invention is suitable for filling product and in particular bulk materials into bags and comprises at least one bag supply, at least one filling module with at least one filling nozzle and at least one bag removal. In this case, the bag supply and the bag removal are arranged in a line and the filling module is arranged withdrawn from the line.

Here in particular, a T-shaped arrangement of the individual assemblies or modules with respect to one another is provided.

Thus, in particular the bag supply and the bag removal are arranged in a line.

As a result of such an arrangement it is preferably possible to assign each arbitrary module to the remaining components or assemblies of the packing machine without necessitating major modification work or complex configurations. In addition, a flexible and nevertheless particularly small packing machine is provided as a result of the T-shaped arrangement.

The bag supply is in particular a module or an assembly which preferably provides pre-fabricated bags e.g. in a magazine or on a stack of bags. Depending on the configuration, however, the bag supply can also comprise a bag production or be provided by a bag production.

Particularly preferably the packing machine is suitable and configured to fill open bags with bulk material.

It is self-evident that even if the application is described with reference to the filling of bulk materials, the filling of pasty and/or liquid products in bags similar thereto with the packing machine according to the invention is also possible or also covered by the protection.

The bag removal preferably comprises at least one conveyor belt by means of which a filled and closed bag can be removed. In particular, at least one closure station is provided for closing the filled bags, which is preferably assigned to the bag removal.

The packing machine according to the invention offers many advantages. A considerable advantage is that the T-shaped arrangement of the filling module with respect to the remaining components or assemblies of the packing machine can provide a particularly small and compact packing machine wherein this can be flexibly selected as required as a result of the adjacent arrangement of the filling module. In particular, standard assemblies can be used so that a filling module provided can also be integrated in the packing machine according to the invention.

Preferably the filling module can also be decoupled and removed from the other assemblies or modules without major expenditure so that the maintenance expenditure is reduced.

In known in-line installations, i.e. installations in which all the modules are arranged in a line, it can be difficult for reasons of space to arrange the filling technology, in particular in configurations with an air filling module, in a line. In addition, the operability or maintainability would be made difficult. This is substantially improved by the packing machine or arrangement according to the invention.

Preferably at least one transfer device is provided in the line between the bag supply and the bag removal, wherein the filling module in particular is arranged adjacent to the transfer device.

Particularly preferably the transfer device is suitable and configured to supply an empty bag from the bag supply to the filling nozzle and remove the filled bag from the filling nozzle and supply it to the bag removal.

Preferably the transfer device comprises at least a first gripper device in order to supply empty bags from the bag supply to the filling nozzle and at least one second gripper device in order to remove full bags from the filling nozzle and supply them to bag removal. As a result of such an arrangement, the handling of empty and full bags can be reliably achieved by means of the gripper devices.

Preferably the filling module can be decoupled and removed from the other assemblies or modules without major expenditure so that maintenance expenditure is reduced.

Particularly preferably the first gripper device is provided for transport of the empty bags and the second gripper device is provided for transfer of the full bags to different sides of the transfer device. As a result of such a configuration, it is possible that different handling steps of the transfer device are executed in parallel. In particular, it is possible to work simultaneously on different sides of the transfer device with the result that it is possible on the one hand to take an empty bag from the bag supply and at the same time remove a full bag from the filling nozzle and/or hang an empty bag onto the filling nozzle and supply a full bag to bag removal.

In expedient further developments to this end, the first gripper device for transport transfer of the empty bags and the second gripper device for transfer of the full bags are arranged offset with respect to one another between 60° and 120°, preferably between 80° and 100° and in particular 90° with respect to one another on a transfer device. The parallel handling of full and empty bags is made possible as a result. Depending on the configuration, different angles with respect to one another can advantageously be selected if, for example, the bag supply and the bag removal are arranged in an angled arrangement with respect to one another which does not form a straight line.

In preferred embodiments the transfer device is provided to be rotatable and/or pivotable at least in sections. In this case, the transfer device comprises at least one rotatable element or a type of pillar on which, in particular the gripper devices are arranged. It is thus possible that by rotation of the transfer device or the pillar it is possible to transfer empty bags from the bag supply to the filling nozzle and full bags from the filling nozzle to the bag removal. This is achieved in particular by the to and fro displacement or rotation of the transfer device or the pillar.

Particularly preferably the transfer device or at least a part thereof is rotatable between 60° and 120°, preferably between 80° and 100° and in particular by 90°. It is thus possible that by rotating to and fro, e.g. by 90° in each case, a bag can be conveyed out from the line and in again by 180° by the gripper devices arranged on different sides. This is achieved whereby the first gripper device is initially rotated by 90° in one direction and attaches the empty bag to the filling nozzle. The transfer device then rotates back so that the second gripper device comes in contact with the then filled bag on the filling nozzle. As a result of rotating back again by 90°, the bag is supplied to bag removal. Smaller or larger angles can also advantageously be provided here if, for example, the bag supply and the bag removal are in an angled arrangement with respect to one another which does not form a straight line.

Particularly preferably the transfer device is configured to be displaceable at least in sections. In this case, displaceable is to be understood in particular also as movable, wherein preferably a longitudinal and/or transverse displacement is possible. Thus, it is in particular possible that the transfer device travels on rails or is suspended on rails and/or is configured to be movable in a different manner in order to pull this out transversely to the arrangement or line of bag supply and bag removal or however to move this in the direction of or along the line of the arrangement of bag supply and bag removal. In this case, the transfer device can be pulled out with respect to the linear arrangement in order to provide sufficient access to the filling module, in particular for maintenance work. By pulling out the transfer device, sufficient space is provided for a person to step between transfer device, bag supply, bag removal and filling module or filling modules so that all the assemblies are readily accessible. In particular, if two filling modules are arranged next to one another which are served by the transfer device as required, the transfer device can be provided to be shiftable or displaceable or movable along the line or in the direction from the bag supply to the bag removal or conversely in order to always arrange this ahead of the filling module which is not to be maintained straight away or which is to be used straight away.

Preferably the first gripper device is provided to be displaceable for the transfer of the empty bags. In this case, the gripper device can preferably be moved forwards and backwards at least in sections in order to take an empty bag from the bag supply and bring this in contact with the filling nozzle after rotation of the transfer device. In this case, the first gripper device can be configured, for example, to be pivotable forwards and back again or displaceable or also telescopable.

In expedient further developments, the first gripper device comprises two gripper arms which in particular can be displaced relative to one another. In this case, these gripper arms can in particular be brought together and moved apart in order to receive an open empty bag from the bag supply and transport or transfer it open in the direction of the filling module in order to hang the open bag on the filling nozzle.

In advantageous further developments the second gripper device for the transfer of full bags is provided to be displaceable, at least in sections. In this case, the second gripper device can in particular be displaced forwards and backwards. To this end, for example, a parallelogram-shaped configuration of the second gripper device can be provided, for example.

Preferably the second gripper device comprises at least one clamping device. In this case, such a clamping device can be provided in particular to remove the full bag from the filling nozzle and in this case in particular to press the still-open bag walls against one another so that the bag walls rest on one another for the subsequent closure.

Preferably the second gripper device is assigned at least one cleaning device. In this case, the cleaning device is used here for cleaning the area of the filled bag on which the top seam is provided subsequently.

The cleaning device preferably comprises at least one blowing strip having at least one first nozzle device and at least one second nozzle device through which at least one air stream can be guided in each case at least temporarily. Preferably the first nozzle device has a fixed blowing-out direction, wherein the second nozzle device also has a fixed blowing-out direction. In this case, the blowing-out directions of the first nozzle device and the second nozzle device differ from one another. Furthermore, preferably at least one control device is provided which is suitable and configured to control the at least one first nozzle device and the at least one second nozzle device at least temporarily in a temporally staggered manner.

A nozzle device can in this case preferably be provided by any known and expediently usable type of nozzle or comprise such a nozzle. A nozzle device can, for example, also be provided by a simple opening or hole or comprise an opening or hole through which an air stream or fluid stream can escape. In this case, such an opening can in particular be considered to be slit-like so that with such an elongate nozzle device longer sections can also be cleaned in one piece with air.

As a result of the different blowing-out directions of the first and the second nozzle device and depending on the configuration of each further nozzle device and the temporally staggered or different control of the nozzle devices via a component with static nozzle devices or nozzles, in particular the air movement of a moving nozzle or an oscillating air jet can be simulated. Thus, an effective cleaning of a working section can be ensured without the need to use moving parts which are more expensive to manufacture and maintain.

As a result of such a configuration of the cleaning device, it can be achieved that during cleaning of the bag mouth edge before closure or welding of the filled bag by the moving air stream, the bag walls or film layers lying on top of one another are opened by the acting air jet so that the air jet impinges upon the inner surfaces with the result that the adhering dust or dirt particles are removed. By varying the air jet preferably with a suitable configuration of the bag, the layers of the side gussets are separated from one another so that this area is also cleaned. The individual film layers are opened quasi successively by the moving or oscillating air stream and cleaned by the thus penetrating air stream.

Additionally, in addition to simulating a simple oscillating movement of an air stream, more complex air streams can be produced in simple and/or complex wave movements over the section to be cleaned.

The different types of nozzle devices can be controlled separately and/or jointly in groups in order to produce suitable air movements.

In particular, the individual nozzle devices can be arranged in rows along the blowing strip wherein the individual rows or groups of nozzle devices of different rows can be acted upon separately with air in order to achieve an advantageous air stream or an advantageous air movement.

Particularly preferably the bag supply, the filling module, the bag removal and/or the transfer device are arranged in at least one housing device at least in sections. In this case, preferably a common cover or housing device is provided. However, depending on the configuration the modules can preferably also be arranged individually or also in groups in separate housing devices. In this case, such housing devices are in particular configured to be substantially dust- and/or airtight so that preferably a hermetic sealing of the entire packing machine is made possible.

Preferably at least two filling modules are provided. In this case, it is provided in particular that only one filling module is used in each case. In particular, different properties can be provided at the filling module so that, for example, a fast product change is possible on a packing machine. To this end, it is provided that the transfer device is movable or displaceable at least in the longitudinal direction of the packing machine or in the line of bag supply and bag removal so that the transfer device can be moved in front of the corresponding filling module to be used. To this end, the filling modules are in particular arranged next to one another.

If at least two filling modules are provided next to one another, the transfer device can preferably be configured and adapted to accept empty bags from the bag supply and to feed full bags to the bag removal, regardless of the filling module in front of which the transfer device is currently located. For this purpose, in particular, the distance between transfer device and bag supply and bag removal can be bridged regardless of the position of the transfer device, for example, by a correspondingly displaceable gripper device. Preferably both filling modules can also be operated alternately, for example, by attaching an empty bag to the first filling module and removing a filled bag from the second filling module.

The method according to the invention is suitable for operating a packing machine as described previously. In this case, a bag from the bag supply is fed out of the line of bag supply and bag removal to the filling module and after filling is guided back into this line and fed to the bag removal.

The method according to the invention also offers the advantages such as have already been described previously for the packing machine. In particular, by the method according to the invention it is possible to provide a packing machine arranged in a T shape, wherein in particular an arbitrary filling module can be arranged next to the remaining assemblies or modules of the packing machine arranged in a line.

Preferably, the method is characterized by the following steps in a suitable number, suitable order and/or combination:

a) picking up an empty bag by the first gripper device (**5**)
b) turning the transfer device (**4**) until the first gripper device (**5**) is in operative connection with the filling module (**100**)
c) transferring the empty bag to the filling nozzle (**101**)
d) turning the transfer device (**4**) until the first gripper device (**5**) is in operative connection with the bag supply (**2**)
e) turning or turning back the transfer device (**4**) until the second gripper device (**6**) is in operative connection with the filling module (**100**) or with the filled bag
f) Picking up the full bag by the second gripper device (**6**)
g) Turning the transfer device (**4**) until the second gripper device with the full bag is in operative connection with the bag removal
h) Handing over the bag to the bag removal.

Steps a) and f), b) and g), c) and h) and/or d) and e) are preferably carried out in parallel or simultaneously. It is in particular preferred in this case that at least temporarily a new empty bag is picked up by the first gripper device and transferred to the filling nozzle when a full bag is removed from the filling nozzle by the second gripper device and transferred in the direction of the bag removal. In this way, the packing machine can be operated in parallel and thus consequently with a higher performance.

Particularly preferably, at least two filling modules are provided, wherein the transfer device is moved in front of the filling module, which is to be used directly. As already described previously, it is possible, for example, to make a quick product change on a machine since different filling modules can be installed in parallel on the machine. Alternatively, it is preferable to work with the same filling modules and the same product in order to increase the overall performance of the packing machine, in particular when a product is filled, which requires a longer residence time in each case for compaction or venting at the filling nozzle.

A further method according to the invention is suitable for maintaining a packing machine such as has been described previously. In this case, the transfer device is moved to release at least the filling module. As a result of the transfer device method a user or maintenance staff are provided with sufficient space in the packing machine between the various assemblies, which enables a particularly simple and therefore also reliable and time-saving maintenance of all components. In addition to the easier access to the filling module, all the other assemblies, in particular the bag supply and the bag removal and the side of the transfer device facing the filling module can be reached more easily.

Particularly preferably, the transfer device is pulled out transversely to the arrangement or line of bag supply and bag removal. This is particularly advantageous if only one filling module is used and a very compact arrangement of the individual assemblies is provided.

In a particularly preferred embodiment, two filling modules are provided, wherein the transfer device is moved in the line between bag supply and bag removal or longitudinal direction. In this case, the transfer device is in particular moved in front of the filling module which is currently not to be maintained and/or which is just to be used for filling.

Further advantages and features of the present invention are obtained from the exemplary embodiment, which is explained hereinafter with reference to the appended figures.

BRIEF DESCRIPTION OF THE DRAWINGS

In the Figures

FIG. **1** shows a purely schematic diagram of an exemplary embodiment of a packing machine according to the invention in a perspective view;

FIG. **2** shows a further purely schematic diagram of an exemplary embodiment of a packing machine according to the invention in a perspective view;

FIG. **3** shows a next purely schematic diagram of an exemplary embodiment of a packing machine according to the invention in a perspective view;

FIG. **4** shows a purely schematic diagram of an exemplary embodiment of a packing machine according to the invention in a view from the front;

FIG. **5** shows a purely schematic diagram of an exemplary embodiment of a packing machine according to the invention in a view from the side of the bag removal;

FIG. **6** shows a further purely schematic diagram of an exemplary embodiment of a packing machine according to the invention in a view from the side of the bag removal;

FIG. **7** shows a next purely schematic diagram of an exemplary embodiment of a packing machine according to the invention in a view from the side of the bag removal;

FIG. **8** shows another purely schematic diagram of an exemplary embodiment of a packing machine according to the invention in a view from the side of the bag removal;

FIG. **9** shows a purely schematic diagram of an exemplary embodiment of a packing machine according to the invention in a view from the side of the bag supply;

FIG. **10** shows a purely schematic diagram of an exemplary embodiment of a packing machine according to the invention in a view from the side of the bag removal;

FIG. **11** shows a further purely schematic diagram of an exemplary embodiment of a packing machine according to the invention in a view from the side of the bag removal;

FIG. **12** shows the view according to FIG. **11** with pulled-out transfer device;

FIG. **13** shows a purely schematic diagram of an exemplary embodiment of a transfer device;

FIG. **14** shows an enlargement according to FIG. **13**;

FIG. **15** shows a further purely schematic diagram of an exemplary embodiment of a transfer device;

FIG. **16** shows an enlargement according to FIG. **15**;

FIG. **17** shows a perspective view according to the view from FIG. **15**;

FIG. **18** shows an enlargement according to FIG. **17**;

FIG. **19** shows a purely schematic view of an exemplary embodiment of a transfer device with a cleaning device according to the invention;

FIG. **20** shows a purely schematic diagram of a cleaning device according to the invention in a perspective view;

FIG. **21** shows a further purely schematic diagram of a cleaning device according to the invention in a perspective view;

FIG. **22** shows a next purely schematic diagram of a cleaning device according to the invention in a perspective view;

FIG. **23** shows another purely schematic diagram of a cleaning device according to the invention in a perspective view;

FIG. **24** shows a further purely schematic diagram of a cleaning device according to the invention in a perspective view;

FIG. **25** shows a purely schematic diagram of an exemplary embodiment of a transfer device and a bag removal in a front view;

FIG. **26** shows a further purely schematic diagram of an exemplary embodiment of a transfer device and a bag removal in a front view;

FIG. **27** shows a next purely schematic diagram of an exemplary embodiment of a transfer device and a bag removal in a front view;

FIG. **28** shows sections from FIGS. **26**, **27** and **28**;

FIG. **29** shows a next exemplary embodiment of a packing machine according to the invention in a perspective view;

FIG. **30** shows the exemplary embodiment according to FIG. **29** in a front view;

FIG. **31** shows the exemplary embodiment according to FIG. **29** in a rear view;

FIG. **32** shows the exemplary embodiment according to FIG. **29** in a plan view; and

FIG. **33** shows the exemplary embodiment according to FIG. **29** in a side view.

DETAILED DESCRIPTION

FIG. **1** shows a packing machine **1** according to the invention in a perspective view. In this case, the packing machine **1** in the exemplary embodiment shown here comprises a bag supply **2**, a bag removal **3** and an interposed transfer device **4**. The bag supply **2**, the bag removal **3** and the transfer device **4** are in this case arranged in a line **12**.

A filling module **100** is provided, pulled out of line **12** and arranged in the exemplary embodiment shown here next to the transfer device **4**. In the exemplary embodiment shown here, empty bags from the bag supply **2** are fed to the filling module **100** by means of the transfer device **4**. The bag thus supplied to the filling module **100** is attached to the filling nozzle **101** of the filling module **100** and filled and guided by the transfer device **4** back into the line **12** and fed to the bag removal **3**.

By arranging the filling module **100** next to the line or pulling out from the line, almost any filling module **100** can be used together with the modules or assemblies of the packing machine **1** arranged in the line **12**. In particular, standard modules can also be used so that no special configuration of a filling module **100** must be provided for the packing machine according to the invention, preferably modules for filling open bags of any filling technology.

In the exemplary embodiment shown here, the packing machine **1** is provided encapsulated, for which the individual assemblies or modules are provided in housing devices **11**, **102**. In this case, in the exemplary embodiment shown here, the assemblies arranged in line **12**, namely the bag supply **2**, the bag removal **3** and the transfer device **4** are provided in a common housing device **11**. In the exemplary embodiment shown here, the filling module **10** is provided in its own housing device **102**.

By arranging the individual assemblies or modules in housing devices **11**, **102**, it is possible to hermetically seal off the filling process depending on the configuration so that hazardous or sensitive substances can also be filled by means of the packing machine **1**. In particular, in the packing machine **1** or in the housing devices **11**, **102** a vacuum can also be applied or a suction can take place so that product which escapes during filling does not escape into the environment but can be safely disposed of or even recycled.

In the exemplary embodiment shown here the bag supply **2** also comprises a bag production **13**, wherein bags are produced for this purpose starting from a film supply **14** and made available to the transfer device **4**.

In the exemplary embodiment shown here the bag removal **3** comprises a closing device **15**, by means of which the filled and still open bags can be closed and a conveyor belt **16** for the removal of the filled and closed bags.

Since the bag supply **2**, the bag removal **3** and the transfer device are arranged in a line **12** and the filling module **100** is arranged pulled out of this line **112**, this results in a T-shaped arrangement of the packing machine **1** according to the invention.

As already mentioned previously, this offers many advantages. An advantage is that by juxtaposing the filling module **100** or by pulling out the filling module **100** from the line **12**, almost any (standard) filling module can be used. As a result, on the one hand, existing filling modules **100** can be used. On the other hand, due to the juxtaposition a filling module can also be changed or exchanged quickly without too much effort. In known inline systems, i.e., systems in which everything is arranged in a line, for reasons of space it can be difficult to arrange the filling technology, in particular the air filling module **105** with pressure chamber **106** shown in the exemplary embodiment, in line. In addition, operability or maintainability would be more difficult.

Another advantage is that due to the T-shaped configuration or arrangement of the packing machine a particularly compact design is possible.

For better illustration of the packing machine **1**, FIGS. **2** to **4** show two perspective views and a front view of the packing machine **1**, wherein in the housing devices **11**, **102**, the housing covers **17**, **103** are not shown to provide overview views of the interior of the packing machine **1** or of the various modules or assemblies.

In FIGS. **5** and **6**, the view of the transfer device **4** from the side of the bag removal **3** is shown once in an overview view and once in an enlarged diagram.

In this view, it can be seen that the filling module **100** is provided or arranged next to the transfer device **4**, wherein the filling nozzle **101** of the filling module **100** can also be identified in this view. Here the inlet hopper **104** of the filling module **100** can also be seen, which is connected to a silo for product to be filled, which is located thereabove but not shown. In this case, product is fed via the inlet hopper **104** into the pressure chamber **106** inside the filling module **100** or here the air filling module **105** to the filling nozzle **101**.

In this view it can be seen that the transfer device comprises a first gripper device **5** and a second gripper device **6**, which are mounted on different sides **7**, **8** of the transfer device **4** or on a base body **18** of the transfer device **4**.

It is shown here that the base body **18** of the transfer device **4** is configured in a pillar-like manner here and is provided in a suspended manner in the exemplary embodiment shown here. In the exemplary embodiment shown, the base body **18** is suspended on rails **19**, whereby the base body **19** with the gripper devices **5**, **6** can be pulled out in this view to the left from the line **12**. This displaceability of transfer device **4** will be shown and described in more detail later.

In addition to the displaceability of the transfer device **4** or the base body **18** of the transfer device **4**, the base body **18** can also be rotated by 90° back and forth in the exemplary embodiment shown here. By turning back and forth the base body **18** of the transfer device **4** and by arranging the first gripper device **5** on one side **7** of the base body **18** and arranging the second gripper device **6** on the side **8** of the base body **18**, by turning the base body **18** back and forth through 90° a bag from the bag supply **2** can be fed by means of the transfer device **4** out from line **12** to the filling module **100** or the filling nozzle **101**. When the base body **18** or the transfer device **4** turns back, the second gripper device **6** comes in contact with the now filled bag at the filling nozzle **101** and can pick this up and can be guided back into the line **12** by turning forwards or re-turning the base body **18** and fed to the bag removal.

In the enlargement in FIG. **6** the first gripper device **5** and the second gripper device **6** can be identified on different sides **7**, **8** of the base body **18** of the transfer device **4**. One of the two gripper arms **9** of the first gripper device **5** can also be identified here.

As can be seen in more detail in the following figures, in the exemplary embodiment shown here the gripper device **5** comprises two gripper arms **9** which can be moved relative to each other so that they serve as a kind of spreader. Thus, it is possible that the gripper arms **9** of the first gripper device **5** pick up an open bag from the bag supply **2** or the bag production and attach this to the filling nozzle **101**. For this purpose, the first gripper device **5** or for this purpose, the gripper arms **9** in the exemplary embodiment shown here can also be displaced forwards and pulled back again, which is shown purely schematically in FIGS. **7** and **8**.

FIG. **9** shows purely schematically the view of the transfer device **4** and the filling module **100** from the direction of the bag supply **2**. In this case, the base body **18** of the transfer device **4** is rotated in such a manner that the second gripper device **6** is aligned in the direction of the filling nozzle **101** of the filling module **100**. In this position, the first gripper device **5** is aligned in such a manner that it can receive an empty bag from the bag supply **2**.

FIG. **10** shows a purely schematic view of the transfer device **4** and the filling module **100** from the side of the bag removal **3**.

Here it can be seen that the second gripper device **6** is provided to be displaceable so that the second gripper device **6** in the exemplary embodiment shown can be pivoted or displaced forwards via a parallelogram-like structure **20** in the direction of the filling nozzle **101**. Thus, the clamping devices **10** of the second gripper device **6** can compress the still open bag walls of the filled bag and then displace the bag back in the direction of the base body **18**.

Subsequently, the base body **18** can be rotated clockwise by 90° here so that the full bag can be displaced back into the line **12** and then fed to the bag removal. At the same time, during the removal of the full bag by the second gripper device **6**, a new empty bag can be picked up by the first gripper device **5**. When turning the full bag forwards into the line **12**, a new empty bag is then simultaneously moved out of line **12** and fed to the filling nozzle **101**.

In FIG. **10** it can also be seen that a cleaning device according to the invention **50** is provided above the second gripper device **6**, which is explained in detail in the subsequent figures.

FIGS. **11** and **12** show purely schematically once again a side view of the packing machine **1** or of the transfer device **4** and the filling module **100** from the direction of the bag removal **3**. When this view is compared, it can be seen that the transfer device **4** or the base body **18** together with the gripper devices **5**, **6** fastened thereon is configured to be movable or extendable.

For this purpose, as already stated in the exemplary embodiment shown here, the base body **18** is suspended in an upper region and displaceable via a rail **19** or a rail system

19. Other technical components or components for displacing the base body **18** can be used analogously.

Since in the exemplary embodiment shown here, the transfer device **4** or the base body **18** of the transfer device **4** can be pulled out transversely to the line **12**, for example, a sufficient space for maintenance staff is provided for maintenance of the packing machine **1**.

Thus, in the extended state of the base body **18** in the exemplary embodiment shown here it is in particular possible to comfortably reach all the assemblies of the packing machine **1**. In particular, the filling module **100** can be reached from the side of the filling nozzle **101**. However, the bag supply **2**, the bag removal and also the otherwise inaccessible side of the transfer device can be maintained and/or repaired.

FIGS. **13** and **14** show the transfer device **4** or the base body **18** of the transfer device **4** of the packing machine **1** in an overview diagram and a detailed view.

The first gripper device **5** and the second gripper device **6** are arranged in this exemplary embodiment offset by 90° on the different sides **7**, **8**. As a result of this arrangement a superimposed operating mode of the empty bag and the full bag transfer or transport is ensured.

A cleaning device **50** is provided above the second gripper device **6**, which comprises a housing device **64**, wherein in the exemplary embodiment shown here a pivotable housing element **65** or a pivotable flap **66** is provided. The cleaning device **50** is explained in more detail in subsequent figures.

In addition, it can be seen that the first gripper device **5** is arranged on the side **7** of the base body **18**, which, in the exemplary embodiment shown here, comprises two gripper arms **9**, which can be moved here relative to each other or pivoted outwards or moved. Thus, the gripper arms act as spreaders and can pick up an empty bag from bag supply **2** or take it from this.

Furthermore, as can already be seen previously in the figures, the first gripper device **5** is provided to be pivotable so that this can be displaced forwards in the direction of the filling nozzle **101** and also in the direction of the bag supply **2**. Depending on the configuration, for example, telescopic gripper arms **9** or other components of the first gripper device **5** can be provided.

In FIGS. **15** and **16**, the base body of the transfer device **4** is rotated in such a manner that the first gripper device **5** is aligned in the direction of the filling nozzle **101** of the filling module **100**. In this position, an open bag not shown is held on the gripper arms **9** in the exemplary embodiment shown here by spreading the two gripper arms **9**, wherein the gripper device **5** is displaced forwards to attach the bag which is held open to the filling nozzle **101**. In addition, the second gripper device **6** with the clamping devices **10** can be identified.

FIGS. **17** and **18** show the side view from FIGS. **15** and **16** again in a slightly perspective view from obliquely above, wherein in addition the cleaning device **50** above the second gripper device **6** is hidden in order to get a better impression of the clamping devices **10** of the second gripper device **6**.

In FIGS. **19** to **24**, the cleaning device **50** according to the invention provided above the second gripper device **6** in the exemplary embodiment shown here and its mode of operation is described in detail.

In this case, the cleaning device **50** according to the invention in the exemplary embodiment shown here is provided for cleaning the so-called head seam area of a bag to be closed by means of blown out air.

In addition to the application of the cleaning device **50** presented here, this can also clean any other work section, in particular in relation to the packing machine **1** described here.

FIG. **19** shows an overview view of the transfer device **4** of the packing machine **1** according to the invention purely schematically, wherein in the exemplary embodiment shown here the cleaning device **50** according to the invention is arranged above the second gripper device **6**.

In this exemplary embodiment, the cleaning device **50** comprises two blowing strips **51** which cannot be seen in FIG. **19**. The cleaning device **50** is provided encapsulated in the exemplary embodiment shown here namely in a housing device **64**. This makes it possible to achieve a hermetic sealing of the blowing strips **51** and the work section to be cleaned. In order to feed the work section to be cleaned or the bag to the cleaning device **50**, the housing device **64** comprises a movable or, in this case, pivotable housing element **65** or a flap **66**, which is shown in the closed state in the view shown.

FIG. **20** shows in a perspective view a cleaning device according to the invention **50**, wherein the movable housing element **65** or the flap **66** is shown in the open state. Thus, the view of the two blowing strips **51** located inside the housing device **64**, shown here as an example, is free.

In FIG. **21** in a perspective view, the assembly of the cleaning device **50** is shown separately in a perspective view. Here also, the flap **66** or the pivotable housing element **65** is shown in the open state, so that the two blowing strips **51** provided in the exemplary embodiment shown here can be seen.

In the exemplary embodiment shown here, the blowing strips **51** each comprise a plurality of first nozzle devices **52**, second nozzle devices **53** and third nozzle devices **54**, which are arranged here in rows **60**, **61**, **62**.

In this case, the three rows **60**, **61**, **62** of the first nozzle devices **52**, the second nozzle devices **53** and the third nozzle devices **54** can be supplied differently with air so that a moving air wave is blown out over the work section to be cleaned.

This is achieved whereby the first nozzle devices **52** or the first row **60** of first nozzle devices **52** each have a first blowing-out direction **55**, which differs from the blowing-out directions **56**, **57** of the other nozzle devices **53**, **54** or rows **61**, **62** of nozzle devices **53**, **54**. In particular, it is preferred and also provided in this exemplary embodiment that the first nozzle devices **52**, the second nozzle devices **53** and the third nozzle devices **54** each have a blowing-out direction **50**, **56**, **57**, which each differ from one another.

Furthermore, a control device **58** not shown in more detail in the figures is provided, which is suitable and configured to control the nozzle devices **52**, **53**, **54** at least at times in a temporally staggered manner

For this purpose, it is provided in the exemplary embodiment shown that for the three rows **60**, **61**, **62** of nozzle devices **52**, **53**, **54** separate air supplies **59** are provided in each case within the blowing strips **51**. This is provided here for both blowing strips **51** and can be identified in FIG. **22** purely schematically by hiding the left blowing strip **51**. By hiding the one blowing strip **51** the separate air supplies **59***a* or the connections for the air supplies **59** of the blowing strip **51** located outside the blowing strip **51** can be seen.

FIGS. **23** and **24** show purely schematically that an extraction device **63** can also be provided to extract the air expelled by the blowing strips **51** or another fluid together with the swirled particles.

For this purpose, in the exemplary embodiment shown here the extraction device **63** is in operative connection with the interior of the housing device **64** via an opening **67** in the housing device **64**.

In the exemplary embodiment shown, a suction pipe **68** is provided, which is connected to this opening **67**. Depending on the configuration, for example, a further suction pipe **68** can be provided on the left side, wherein here the opening **67** is closed on the left side by a cover **69**.

In the exemplary embodiment shown, as has already been described previously, it is provided that a hermetically substantially sealed cleaning space is created by closing the movable housing element **65** or the flap **66** of the housing device **64**. Depending on the application, the extraction device is then set so that at least the air volume or generally the volume which is blown into the housing **64** through the blowing strips **51** is extracted. In particular, it is preferred that at least so much is extracted that at least a small vacuum is created inside the housing device **64**. This is particularly important for systems that are used in dust explosion areas. In non-critical working environments or in general, a non-hermetically closing housing device **64** or even no housing device **64** can be provided.

In FIGS. **23** and **24** it can also be seen once again that the different rows **60**, **61**, **62** of the first, second and third nozzle devices **52**, **53**, **54** are connected via different air supply lines **59***a* per blowing strip **51**.

Thus, the different control of the individual nozzle devices **52**, **53**, **54** or the individual rows **60**, **61**, **62** of these nozzle devices **52**, **53**, **54** is also possible differently per blowing strip **51**. Thus, in the exemplary embodiment shown, an air wave or a moving air stream can be generated over the entire length and also over the width of the work section to be cleaned, whereby in particular the outlet pattern of a pendulum nozzle is modeled without using moving parts. In this case, in the blow-out pattern of a pendulum nozzle it is achieved that when cleaning the bag section for the subsequent head seam, the bag walls to be joined subsequently are blown apart so that product located between the bag walls is removed.

In particular, by means of an oscillating air jet it is achieved that when cleaning the bag mouth edge before closing or welding the filled bag by the moving air flow, the bag walls or film layers located on top of one another are opened by the acting air jet, so that the air jet impinges upon the inner surfaces, whereby the adhering dust or dirt particles are removed here. By changing the air jet, the layers of the side folds are also separated from each other with a corresponding design of the bag, so that this area is also cleaned. The individual film layers are as it were opened one after the other by the moving or oscillating air flow and cleaned by the air flow thus penetrating.

In FIGS. **25** to **28**, the transfer of a filled bag by the transfer device **4** to the bag removal **3** is shown purely schematically.

In this case, a parallelogram-like structure **20** is provided, on which the second gripper device **6** with the clamping devices **10** is provided. This structure **20** is also used to remove the full bag from the filling nozzle **101** or to bring the second gripper device **6** into contact with the filled bag.

In the basic state, the second gripper device **6** abuts relatively closely against the base body **18** of the transfer device **4**. In the following FIGS. **26** and **27** it can be seen how the second gripper device **6** moves forward in the direction of the bag removal **3** by displacing the parallelogram-like structure **20**.

In this case, the closure device **15** can also be seen here, which is configured as a welding station **21** in the exemplary embodiment shown here.

FIG. **28** shows purely schematically by the side-by-side representation of the individual positions of the parallelogram-like structure **20** that due to the special configuration of this parallelogram-like structure **20**, the filled bag is transferred by means of gripper device **6** of the transfer device **4** approximately on a line or approximately rectilinearly to the bag removal **3**.

FIGS. **29** to **33** show a further exemplary embodiment of a packing machine **1** according to the invention. In this case, the configuration of this exemplary embodiment substantially corresponds to the exemplary embodiment already described.

In particular, the bag supply **2**, the transfer device **4** and the bag removal **3** are arranged in a line **12**. In contrast to the exemplary embodiment shown previously, not just one filling module **100** is pulled out of the line **12** and arranged next to the transfer device **4** but two filling modules **100** are arranged side by side outside the line **12**.

In order to select which of the two filling modules **100** is used, in the exemplary embodiment shown here the transfer device **4** is provided to be movable so that this can be arranged either in front of one or the other filling module **100**. For example, a quick product change or maintenance of a filling module **100** can be carried out, for which the transfer device **4** can be pushed or moved in front of the filling module **100** to be used or not to be maintained.

At the same time as the transfer device **4** can be displaced in the line **12**, depending on the configuration, a movability or a displaceability of the transfer device **4** transverse to the line **12** can also be provided as in the exemplary embodiments shown previously.

In order to ensure a reliable picking up of bags from the bag supply **2** and a reliable delivery of the filled bags by the transfer device **4** to the bag removal **3**, the individual components of the transfer device **4** can accordingly be provided to be displaceable.

Depending on the configuration, however, the bag supply **2** and/or the bag removal **3** can also be co-displaced so that there are always optimal distances between the individual assemblies.

While particular embodiments of the present packing machine and method for operation have been described herein, it will be appreciated by those skilled in the art that changes and modifications may be made thereto without departing from the invention in its broader aspects and as set forth in the following claims.

Reference list	
1	Packing machine
2	Bag supply
3	Bag removal
4	Transfer device
5	First gripper device
6	Second gripper device
7	Side
8	Side
9	Gripper arm
10	Clamping device
11	Housing device
12	Line
13	Bag production
14	Film supply
15	Closure device
16	Conveyor belt

-continued

Reference list	
17	Housing cover
18	Base body
19	Rail
20	Parallelogram-like structure
21	Welding station
50	Cleaning apparatus
51	Blowing strip
52	First nozzle device
53	Second nozzle device
54	Third nozzle device
55	Blowing-out direction
56	Blowing-out direction
57	Blowing-out direction
58	Control device
59	Air supply
59a	Air supply
60	Row
61	Row
62	Row
63	Suction device
64	Housing device
65	Housing element
66	Flap
67	Opening
68	Suction pipe
69	Cover
100	Filling module
101	Filling nozzle
102	Housing device
103	Housing cover
104	Inlet hopper
105	Air filling module
106	Pressure chamber

The invention claimed is:

1. A packing machine for filling product into bags, the packing machine comprising:

at least one bag supply, at least one filling module with at least one filling nozzle, and at least one bag removal device,

wherein the at least one bag supply and the at least one bag removal device are arranged in a line and the at least one filling module is horizontally spaced from the line,

wherein at least one transfer device is provided in the line between the at least one bag supply and the at least one bag removal device, the at least one transfer device being configured to feed an empty bag from the at least one bag supply to the at least one filling nozzle and to feed a filled bag from the at least one filling nozzle to the at least one bag removal device,

wherein the at least one filling module is arranged next to the at least one transfer device, and

wherein the at least one bag supply, the at least one transfer device, the at least one bag removal device and the at least one filling module form a T-shaped arrangement and the at least one transfer device transfers the empty bag from the at least one bag supply to the at least one filling module while the at least one filling module is horizontally spaced from the line and in the T-shaped arrangement.

2. The packing machine according to claim **1**, wherein the at least one transfer device comprises at least a first gripper device to feed empty bags from the at least one bag supply to the at least one filling nozzle and wherein the at least one transfer device comprises at least a second gripper device to feed full bags from the at least one filling nozzle to the at least one bag removal device.

3. The packing machine according to claim **2**, wherein the at least one transfer device includes a body, wherein the first gripper device for the transfer of the empty bags and the second gripper device for the transfer of the full bags are arranged on opposing sides of the body.

4. The packing machine according to claim **2**, wherein the first gripper device for the transfer of the empty bags and the second gripper device for the transfer of the full bags are arranged offset with respect to one another by an angle between 60° and 120° on the at least one transfer device.

5. The packing machine according to claim **1**, wherein the at least one transfer device at least in sections is provided to be rotatable and/or pivotable.

6. The packing machine according to claim **5**, wherein the at least one transfer device is rotatable by an angle between 60° and 120°.

7. The packing machine according to claim **1**, wherein the at least one transfer device is configured to be movable at least in sections.

8. The packing machine according to claim **2**, wherein the first gripper device for the transfer of the empty bags is configured to be displaceable.

9. The packing machine according to claim **2**, wherein the first gripper device comprises at least two gripper arms which can be displaced relative to one another.

10. The packing machine according to claim **2**, wherein the second gripper device for the transfer of the full bags is configured to be displaceable.

11. The packing machine according to claim **2**, wherein the second gripper device comprises at least one clamping device.

12. The packing machine according to claim **2**, wherein the second gripper device is assigned at least one cleaning device.

13. The packing machine according to claim **1**, wherein the at least one bag supply, the at least one filling module, the at least one bag removal device and/or the at least one transfer device are arranged at least in sections in at least one housing device.

14. The packing machine according to claim **1**, wherein the at least one filling module comprises at least two filling modules.

15. The packing machine according to claim **14**, wherein the at least two filling modules are arranged side by side.

16. A method for operating a packing machine according to claim **1**, wherein a bag from the at least one bag supply is fed out from the line to the at least one filling module and after filling is guided back into the line and fed to the at least one bag removal device.

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